# UPSC Prelims

# Environment & Ecology

## Class No. 1

Basics of Environment and Ecology

# Class Plan

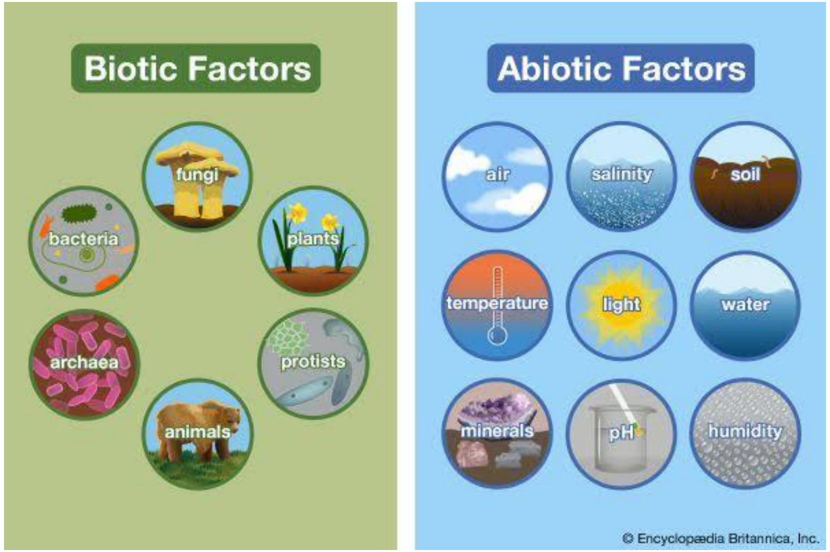
| Class | Date | Day | Topic |
|---|---|---|---|
| 1 | 4/4/2023 | Tuesday | Basics of Environment 1 |
| 2 | 5/4/2023 | Wednesday | Basics of Environment 2 |
| 3 | 6/4/2023 | Thursday | Biodiversity and Indian laws |
| 4 | 7/4/2023 | Friday | Biodiversity : Mapping |
| 5 | 8/4/2023 | Saturday | Biodiversity : Species |
| 6 | 10/4/2023 | Monday | Climate Change |
| 7 | 11/4/2023 | Tuesday | International Conventions 1 |
| 8 | 12/4/2023 | Wednesday | International Conventions 2 |

# Class Plan

| | | | |
|---|---|---|---|
| 9 | 13/4/2023 | Thursday | Sustainable Development |
| 10 | 14/4/2023 | Friday | Sustainable Development and Energy |
| 11 | 15/4/2023 | Saturday | Renewable and Nuclear Energy |
| 12 | 17/4/2023 | Monday | Environmental Pollution 1 |
| 13 | 18/4/2023 | Tuesday | Environmental Pollution 2 |
| 14 | 19/4/2023 | Wednesday | Reserve Class 1 |
| 15 | 20/4/2023 | Thursday | Reserve Class 2 |

# Environment

- The environment is defined as ‘the sum total of living, non-living components; influences and events, surrounding an organism.



# What is Ecology?

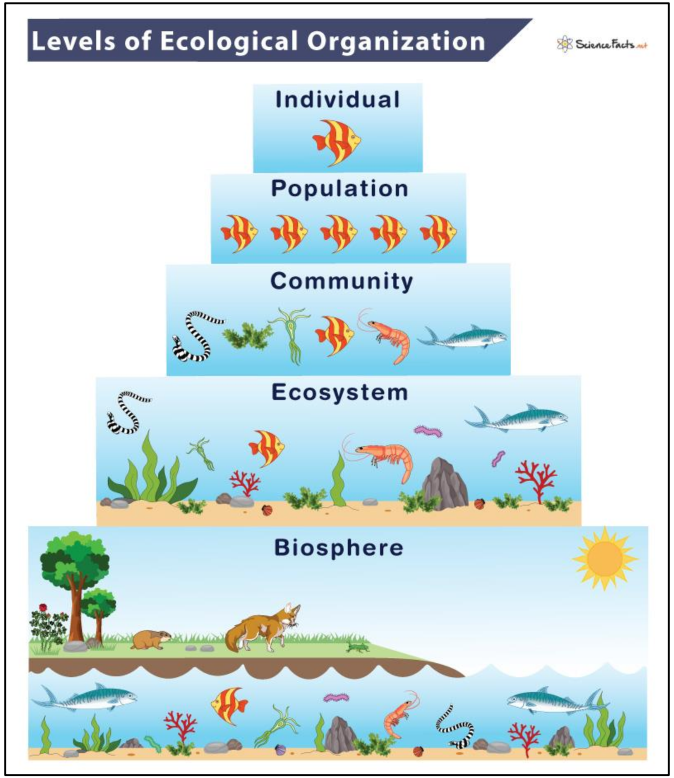
- The study of the interactions between living organisms and their biotic and abiotic environments.
- Therefore, It is the study of the relationship of plants and animals to their physical and biological environment.

# What is Ecology?

# Levels of Organisations In Ecology

- **Organism** is an individual living being that has the ability to act or function independently. It may be plant, animal, bacterium, fungi, etc.
- **Population** is a group of organisms usually of the same species, occupying a defined area during a specific time.
- **Community** refers to all the populations of different species living and interacting in a particular area or habitat. A community may consist of different species of plants, animals, fungi, and microorganisms, which are interdependent and affect each other's survival and well-being.

# Levels of Organisations In Ecology

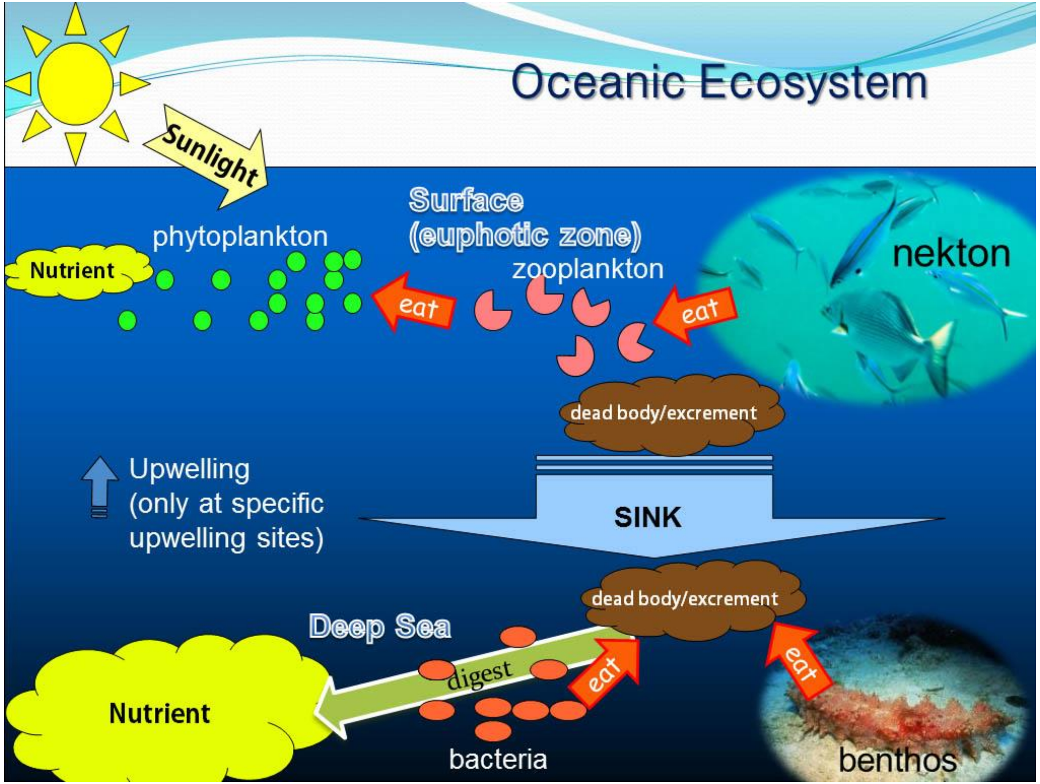


# Levels of Organisations In Ecology: ECOSYSTEM

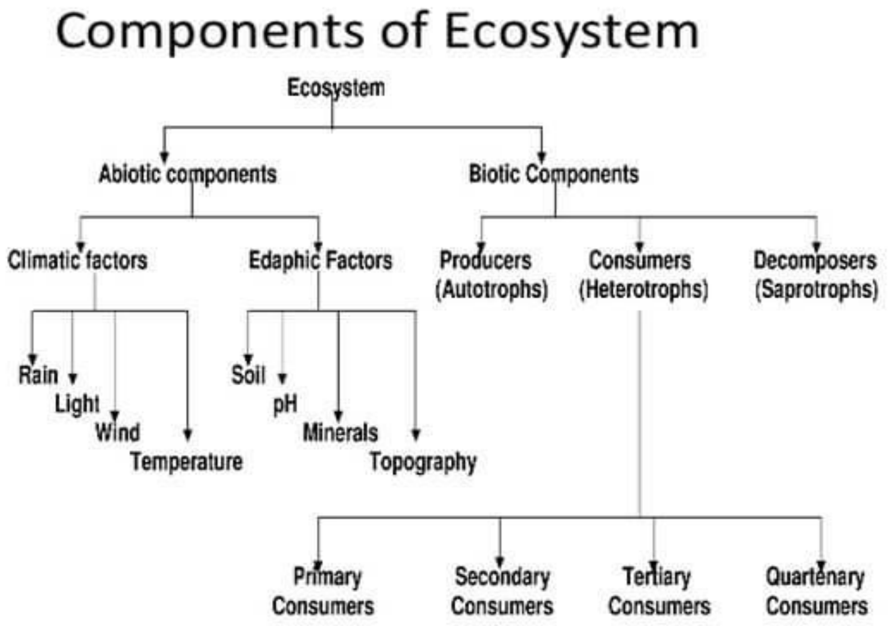


- An ecosystem is defined as a structural and functional unit of biosphere consisting of community of living beings and the physical environment, both interacting and exchanging materials between them.
- The term "ecosystem" was first coined by British ecologist Arthur Tansley in 1935.

# Levels of Organisations In Ecology: ECOSYSTEM



# COMPONENTS OF ECOSYSTEM



# Functional Units of an Ecosystem

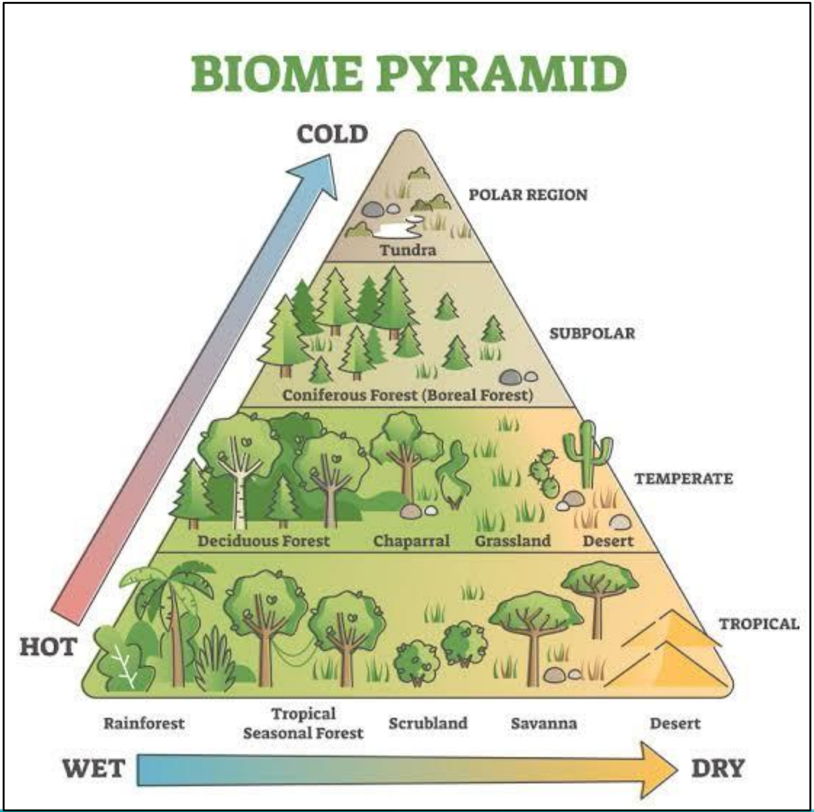
- **Productivity**: It refers to the rate of biomass production.
- **Energy flow**: It is the sequential process through which energy flows from one trophic level to another. The energy captured from the sun flows from producers to consumers and then to decomposers and finally back to the environment.
- **Decomposition** – It is the process of breakdown of dead organic material. The top-soil is the major site for decomposition.
- **Nutrient cycling** – In an ecosystem nutrients are consumed and recycled back in various forms for the utilisation by various organisms.

# Biome



- A biome refers to a large geographical region characterized by a particular set of climate conditions, plant and animal life.
- Each biome has unique characteristics that determine the kinds of plants and animals that can survive and thrive in that particular environment.
- Types of biomes: forests, grasslands, tundra, deserts, and aquatic biomes such as oceans, rivers, and lakes.
- You must read this from GC leong.

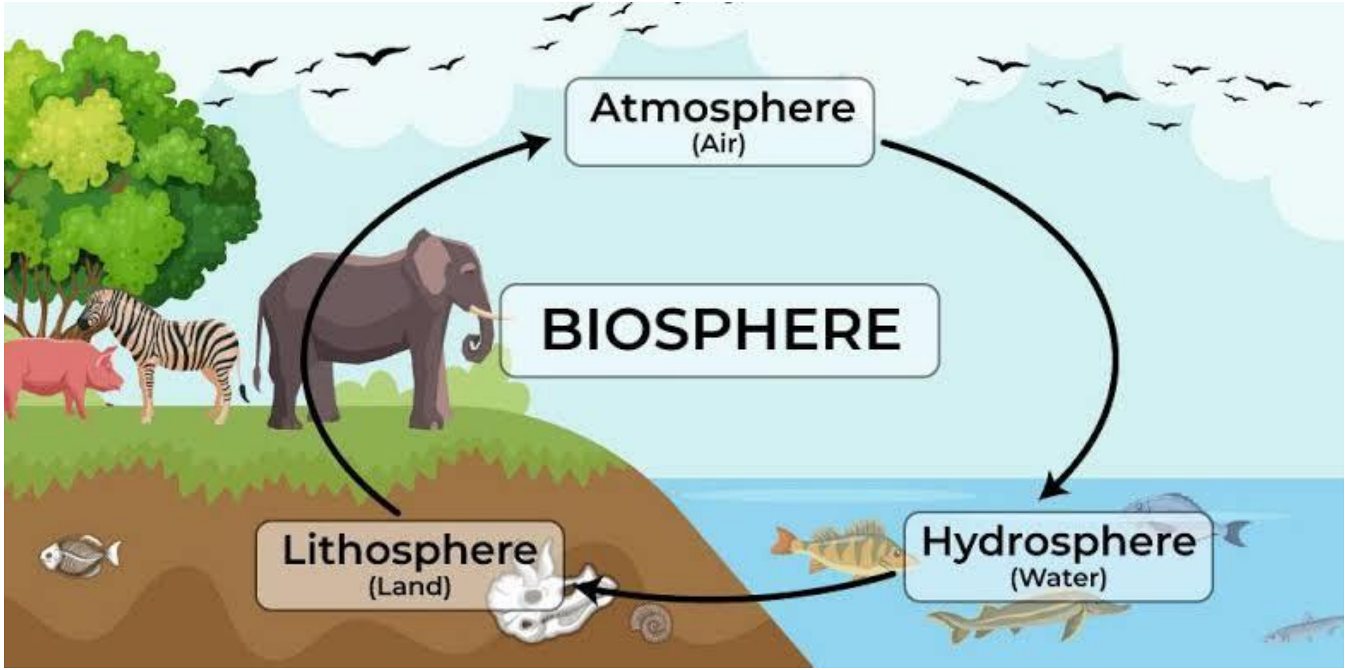
# Biome


BIOME PYRAMID
COLD
HOT
POLAR REGION
Tundra
SUBPOLAR
Coniferous Forest (Boreal Forest)
TEMPERATE
Deciduous Forest
Chaparral
Grassland
Desert
TROPICAL
Rainforest
Tropical Seasonal Forest
Scrubland
Savanna
Desert
WET
DRY

# BIOSPHERE

- Biosphere is a part of the earth where life can exist.
- It represents a highly integrated and interacting zone comprising atmosphere (air), hydrosphere (water) and lithosphere (land).
- Biosphere is approximately 20 km thick. Most life occurs between 500m below the surface of the ocean and about 6 km above the sea level.

# BIOSPHERE



Atmosphere
(Air)
BIOSPHERE
Hydrosphere
(Water)
Lithosphere
(Land)

# HABITAT

- It is the place where an organism or a community of organisms lives, including all living and nonliving factors or conditions of the surrounding environment.
- Microhabitat is a term for the conditions and organisms in the immediate vicinity of a plant or animal.

# HABITAT

# NICHE



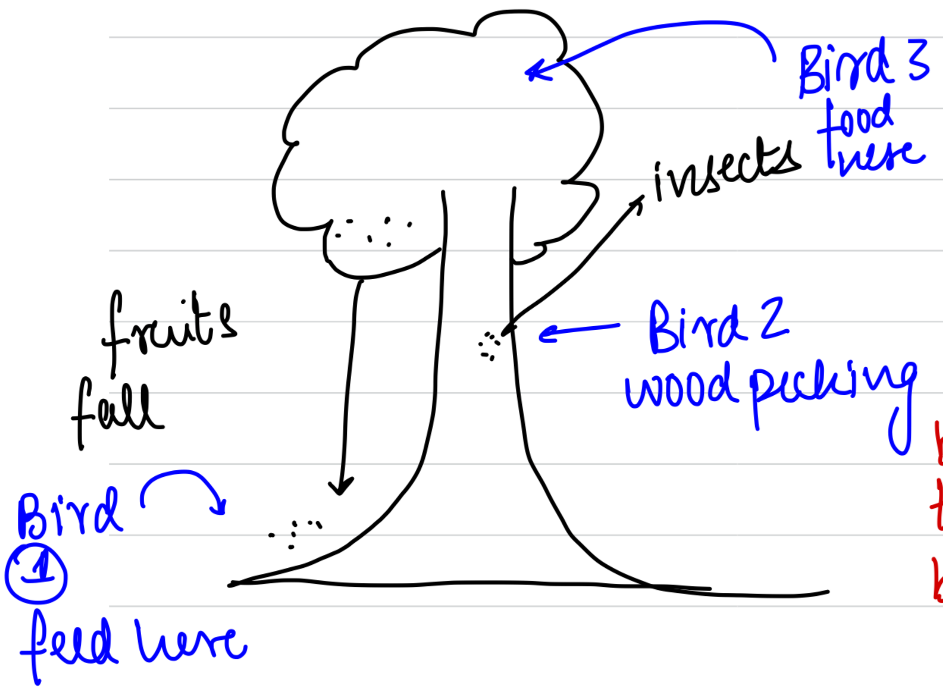
- Niche refers to the unique role or position of a species within an ecosystem, including the physical and biological conditions it requires to survive and reproduce.
- It can be described by the specific set of environmental conditions, such as temperature, humidity, light, soil type, and food availability, that a species requires to survive and thrive.
- The niche of an organism and its interactions is determined by where it stands in the ecological structure of the ecosystem. (Producers, Consumers, Decomposers)

# Habitat and Niche

- Habitat : Physical Space occupied by an organism.
- Niche : Functional Space occupied by an organism where it gets the resources needed to survive.

# Habitat and Niche





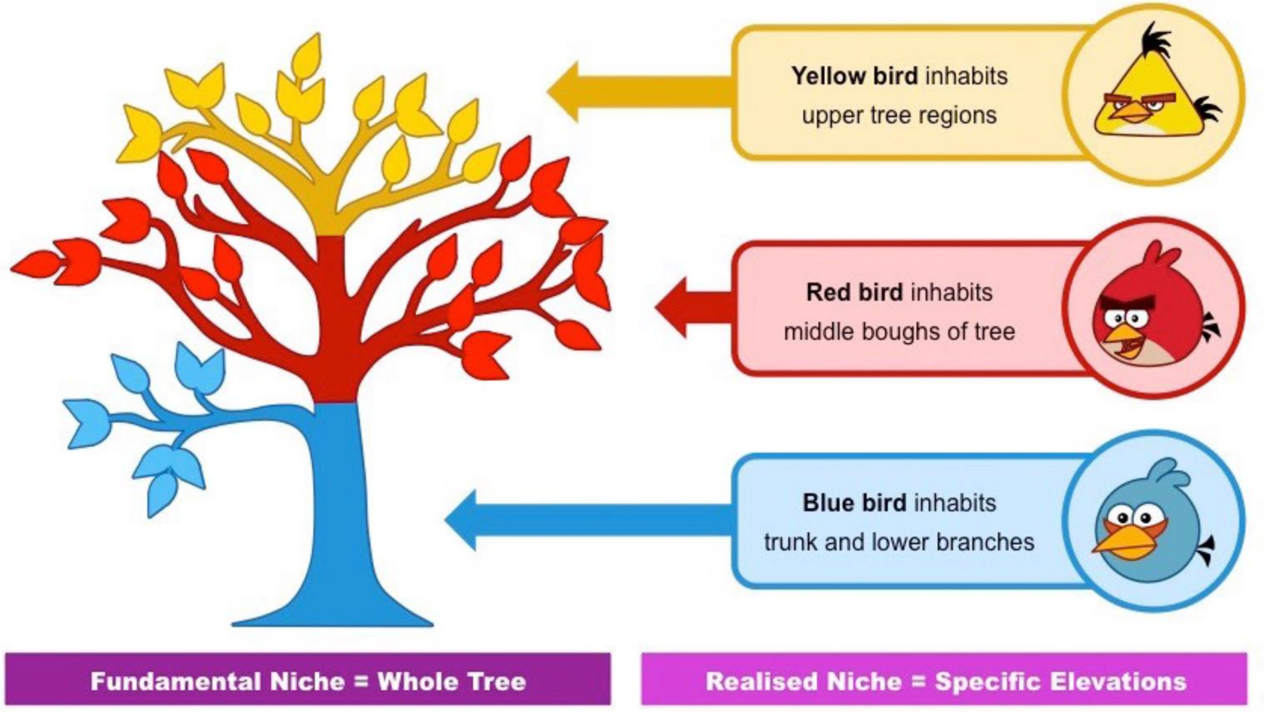

Bird 1, 2, 3
↓
Habitat = tree

Niche = bottom, trunk, branches

# Niche : Types

- Based on the interactions of species, niche are of 3 types :
- Habitat niche : physical area in the the habitat that a species occupies.
- Trophic Niche : Trophic level occupied by the species in the food chain / ecological chain.
- Multidimensional Niche : Fundamental Niche and Limiting factors.
- Fundamental Niche : where an organism could exist without ecological interactions.
- Realised niche : population exists here in the presence of interactions and competition.

# Niche : Types



**Yellow bird** inhabits upper tree regions
**Red bird** inhabits middle boughs of tree
**Blue bird** inhabits trunk and lower branches
Fundamental Niche = Whole Tree
Realised Niche = Specific Elevations

# NICHE OVERLAP

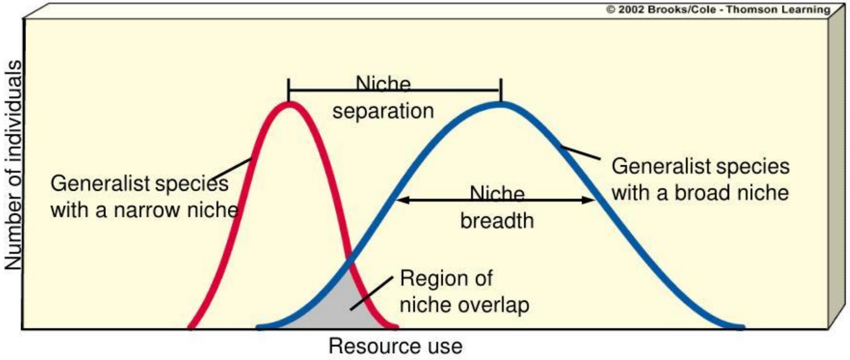
- If 2 organisms have the same niche : there is increased competition, leading to less chances of survival.
- Gauss Law : Competitive exclusion principle : Competition in case of the same niche of 2 species will lead to the exclusion of 1 from that niche.
- Darwin’s finches and galapagos islands.

# NICHE OVERLAP

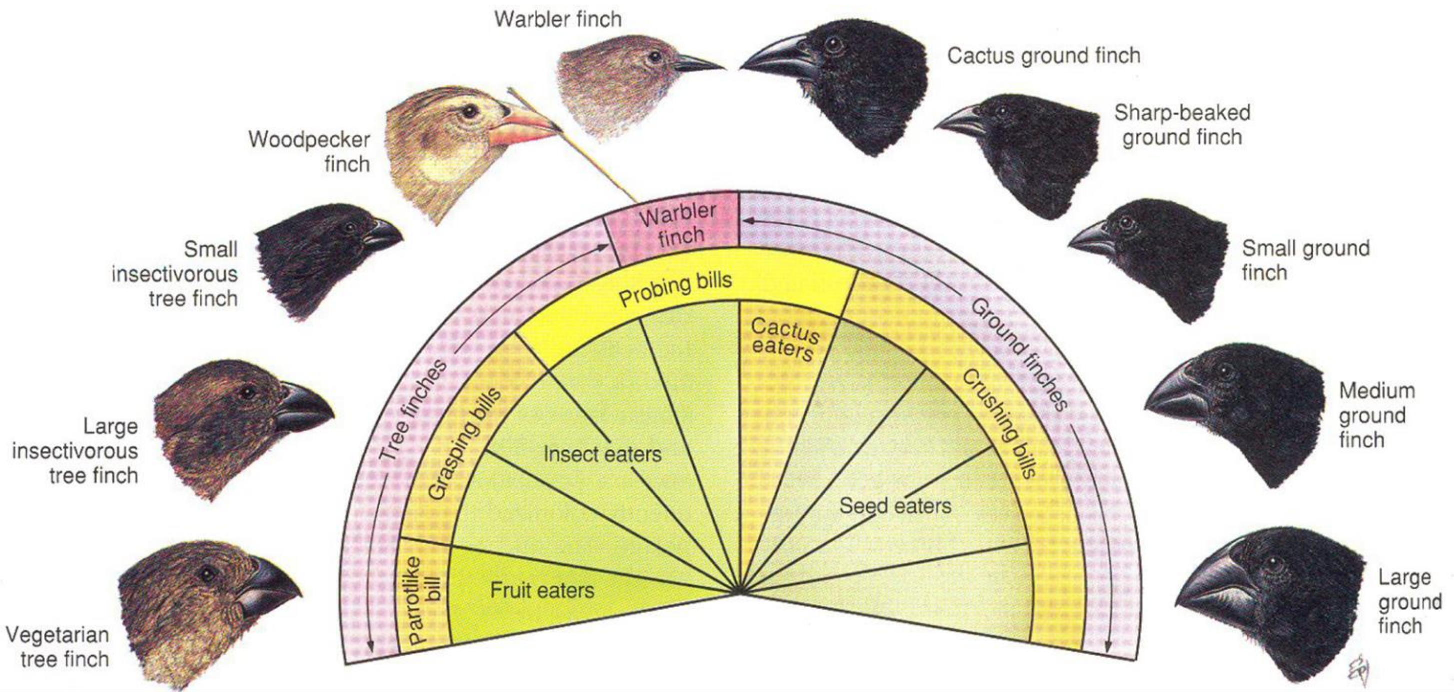
- Niche overlap describes the situation in which co-occurring species share parts of their niche space with each other.
- Overlapping niches may lead to Competitive Exclusion.
  - **Competitive Exclusion:** Two species which compete for the same limited resource cannot coexist at constant population values.
- Niche overlap is reduced by resource partitioning.
  - **Resource partitioning** is the process of moving things around in order to satisfy the niche size to an appropriate level.

# NICHE OVERLAP

## Niche Overlap

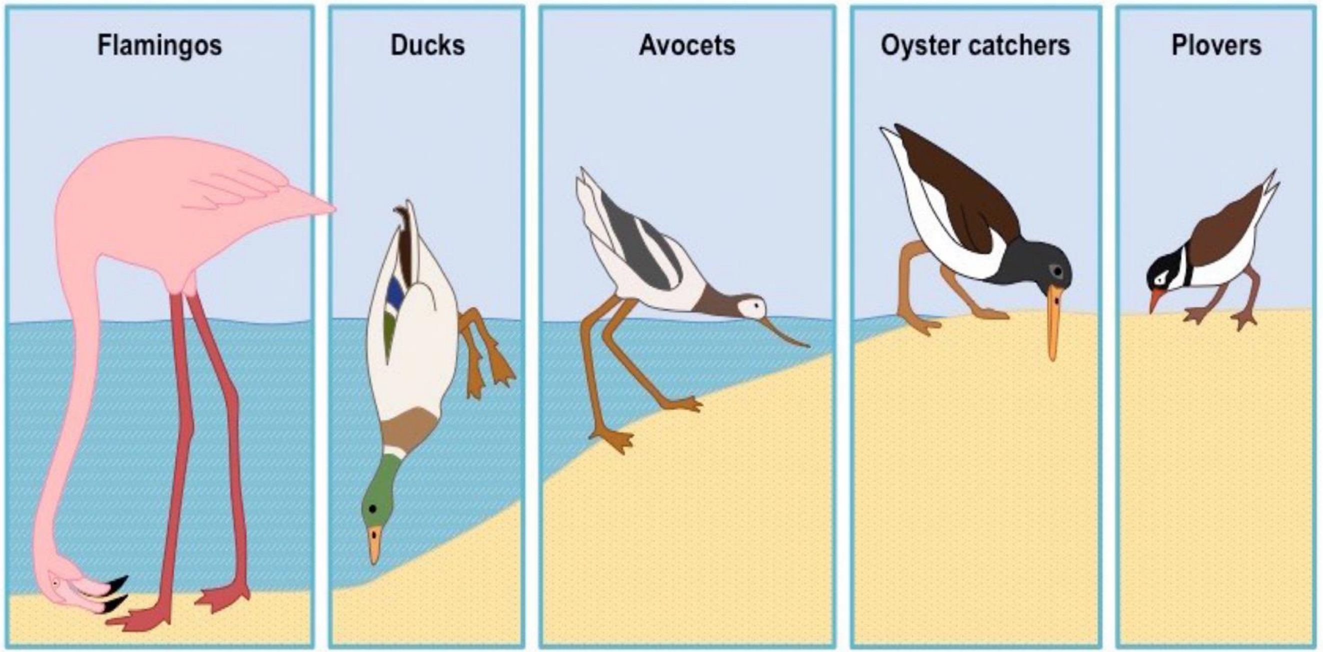


# NICHE OVERLAP : competitive exclusion



# NICHE OVERLAP : Resource Partitioning

**Resource Partitioning:** Species alter their use of the niche to avoid competition, by dividing resources among them



# ECOTONE

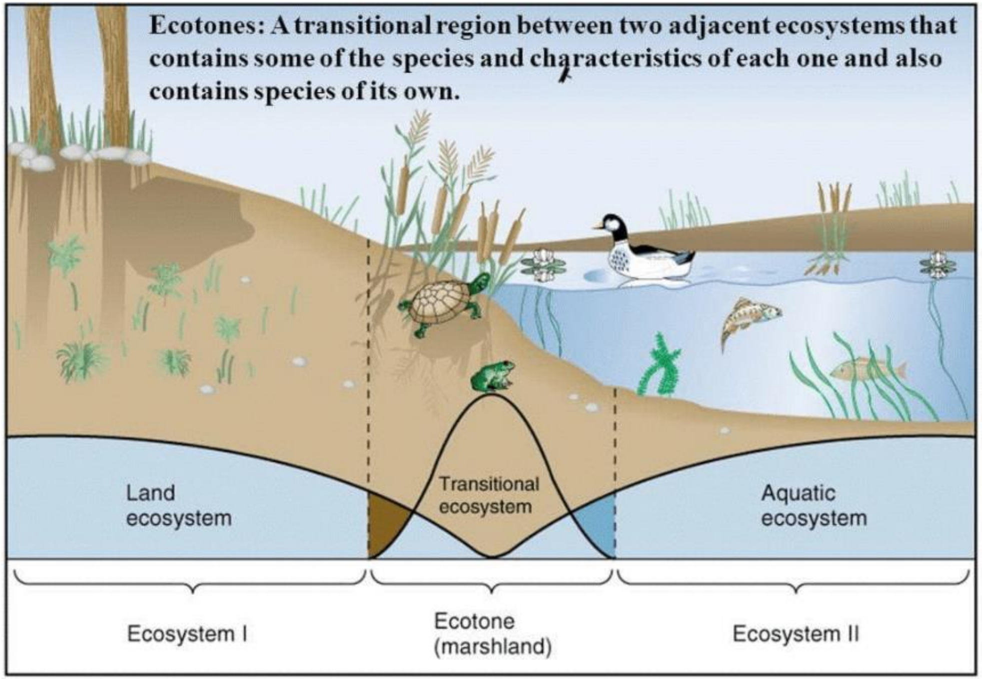
Ecotone is a zone of junction between two or more diverse ecosystems.

- Such areas have richness in biodiversity due to edge effect.

Examples of ecotones include:

- **Marshlands:** Dry and wet ecosystems
- **Mangrove forests:** Terrestrial and marine ecosystems
- **Grasslands:** Desert and forest, and
- **Estuaries:** Saltwater and freshwater

# ECOTONE



Ecotones: A transitional region between two adjacent ecosystems that contains some of the species and characteristics of each one and also contains species of its own.
Land ecosystem
Transitional ecosystem
Aquatic ecosystem
Ecosystem I
Ecotone (marshland)
Ecosystem II

# Characteristics of Ecotone



- It is a zone of tension
- It is linear as it shows progressive increase in species composition of one in coming community and a simultaneous decrease in species of the other outgoing adjoining community.
- A well-developed ecotone contains some organisms which are entirely different from that of the adjoining communities.

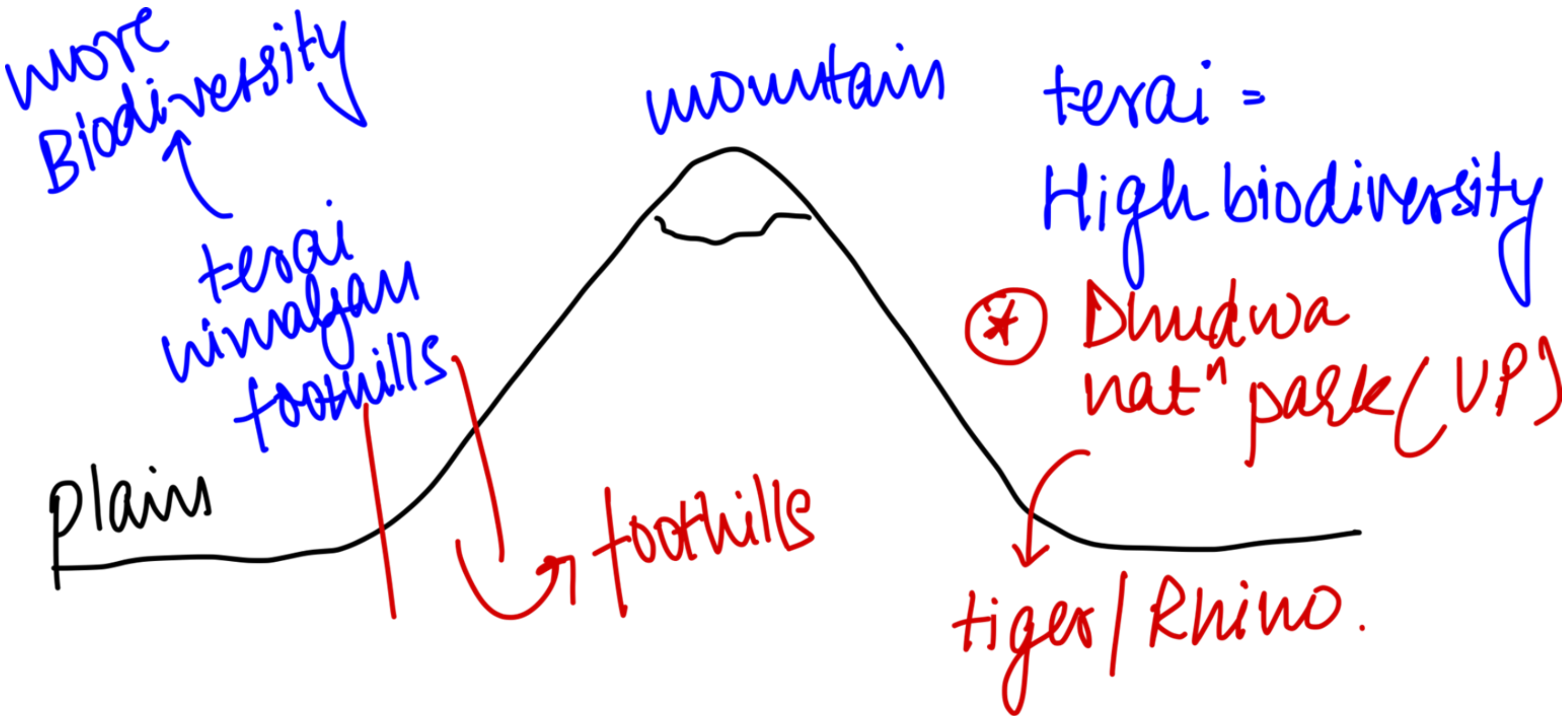
# Examples of Ecotone

- Coral Reefs (we will see this in detail later) : Marine Animals that release limestone (Calcium carbonate). This limestone in the marine environment of the sea lead to edge effect and ecotone presence.
- Wetlands : submerged with water. Midway between land and lakes, therefore have much higher biodiversity. (Ramsar convention : we will see later in detail)

# Edge Effect

- Edge effect refers to the changes in population or community structures that occur at the boundary of two habitats.
- Sometimes the number of species and the population density of some of the species in the ecotone is much greater than either community. This is called the edge effect.
- Species adapted to survive in edge effect areas are called ecotypes.

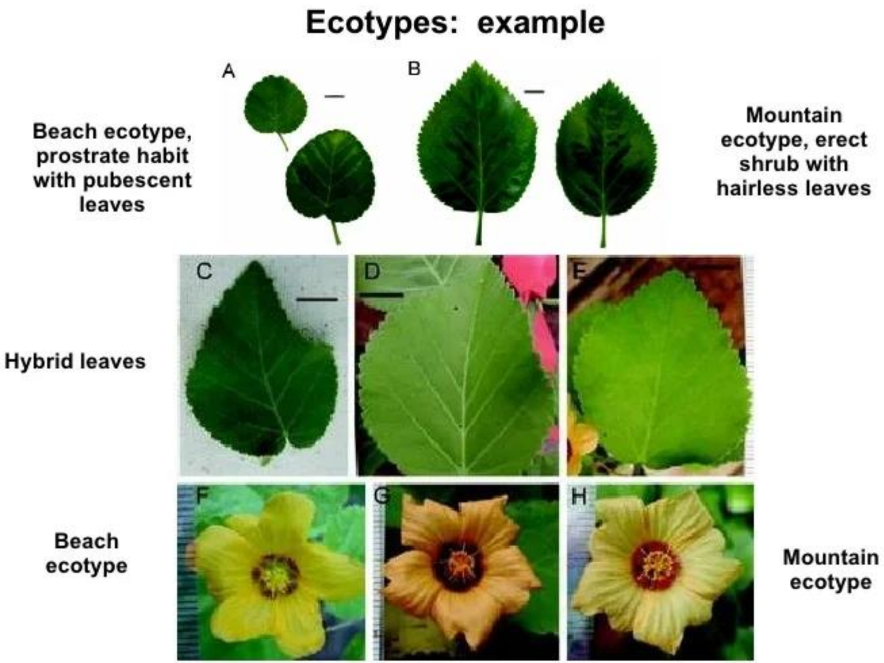
# Edge Effect


more Biodiversity
terai himalayan foothills
mountain
plain
foothills
terai = High biodiversity
Dhudwa natn park (UP)
tiger / Rhino.

# ECOTYPE

- An Ecotype is a population of species that differs genetically from other populations of the same species because local conditions have selected for certain unique physiological or morphological characteristics.
- Ecotypes are adapted to survive in an ecotone.
- Ex : Royal Bengal Tiger : adapted to mangrove, can drink salt water.
- Examples: Kharai Camel (Gujarat), Indian Rhino

# ECOTYPE




Ecotypes of *Sida fallax* and their hybrids in Hawai'i. A. Beach ecotype. B. Mountain ecotype. C, D, and E: hybrid leaves. F: Beach flower. G. Hybrid flower. H. Mountain flower.

# Functional Units of an Ecosystem

- **Productivity**: It refers to the rate of biomass production.
- **Energy flow**: It is the sequential process through which energy flows from one trophic level to another. The energy captured from the sun flows from producers to consumers and then to decomposers and finally back to the environment.
- **Decomposition** – It is the process of breakdown of dead organic material. The top-soil is the major site for decomposition.
- **Nutrient cycling** – In an ecosystem nutrients are consumed and recycled back in various forms for the utilisation by various organisms.

# TROPHIC LEVEL

- A trophic level is the representation of energy flow in an ecosystem.
- It  is the position it occupies in a food chain.
- It deals with how the members of an ecosystem are connected based on nutritional needs.
- The trophic level interaction involves three concepts:
  - Food Chain
  - Food Web
  - Ecological Pyramids

# Energy in the ecosystem

- Plants absorb less that 1% of the sunlight that reaches them.
- Photosynthetic organisms make 170 billion metric tons of food each year.
- 2 processes : photosynthesis and respiration.

# PRIMARY PRODUCERS (AUTOTROPHS)

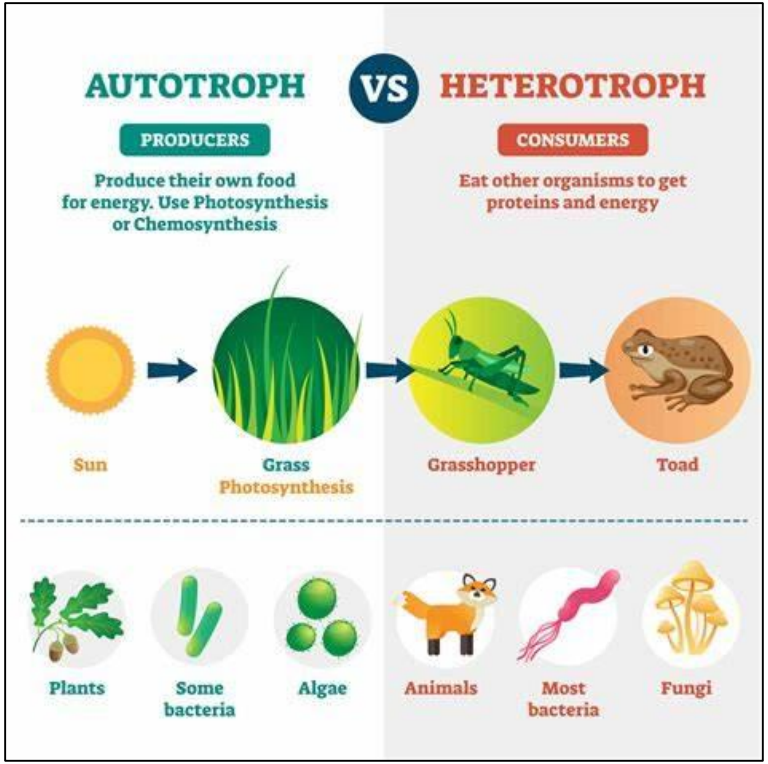
- Word Breakdown : Auto-trophs
- Primary producers are basically green plants (and certain bacteria and algae).
- They synthesise carbohydrates from simple inorganic raw materials like carbon dioxide and water in the presence of sunlight by the process of photosynthesis for themselves, and supply indirectly to other non-producers.
- In terrestrial ecosystem, producers are basically herbaceous and woody plants, while in aquatic ecosystem producers are various species of microscopic algae.

# PRIMARY PRODUCERS (AUTOTROPHS)

- Producers : Phototrophs or chemotrophs
- Phototrophs : organisms that perform photosynthesis and contain chlorophyll

  : Carbon dioxide + water + sunlight = sugar + oxygen
- Chemotrophs : do certain chemical reactions to obtain their food.
- Ex : Sulphur bacteria like thiobascillus.
- Carbon dioxide + water + hydrogen sulphide + oxygen = Carbohydrates + sulphuric acid.

# PRIMARY PRODUCERS (AUTOTROPHS)



# CONSUMERS : Hetero- trophs

- Consumers are incapable of producing their own food (photosynthesis).
- They depend on organic food derived from plants, animals or both.
- Consumers can be divided into two broad groups namely micro and macro consumers.

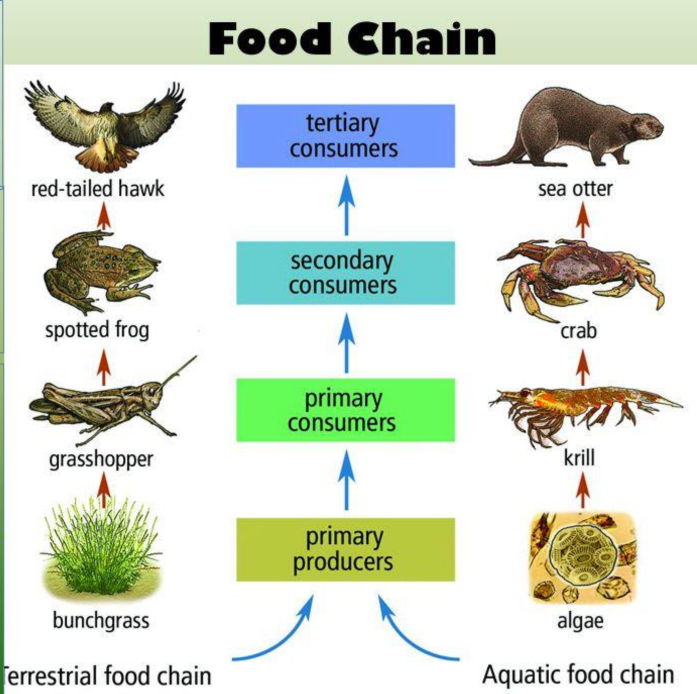
# CONSUMERS : Hetero- trophs

**Tertiary consumers** – eat primary AND secondary consumers

**Secondary consumers** – eat primary consumers

**Primary consumers** – eat primary producers

**Primary producers** – produce the initial food



# MACRO CONSUMERS

- They feed on plants or animals or both and are categorised on the basis of their food sources.
- **Herbivores** are primary consumers which feed mainly on plants e.g. cow, rabbit.
- **Secondary consumers** feed on primary consumers e.g. wolves.
- **Carnivores** which feed on secondary consumers are called tertiary consumers e.g. lions which can eat wolves.
- **Omnivores** are organisms which consume both plants and animals e.g. man, monkey.
- **Scavengers** : feed on dead and decaying organisms

# Scavengers

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

- Scavengers feed on carrion (dead or injured animal corpses)
- Scavengers will feed on these dead plants / animals and decomposers will finish the job.

# Scavengers

CROW
HYENA
FOX
JACKAL
CONDOR
WILD BOAR
TASMANIAN DEVIL
MARABOU STORK
HORNET
WASPS
SEAGULL
VULTURE
OPOSSUM
RACCOON
YELLOW JACKET
ANT
FLESH FLY
DUNG BEETLE

# MICRO CONSUMERS

- They are bacteria and fungi which obtain energy and nutrients by decomposing dead organic substances (detritus) of plant and animal origin.
- They feed on small microscopic bits of dead organic matter and convert them into inorganic nutrients.
- The products of decomposition such as inorganic nutrients which are released in the ecosystem are reused by producers and thus recycled.
- Earthworm and certain soil organisms (such as nematodes, and arthropods) are detritus feeders and help in the decomposition.

# Decomposers and detritivores

| Parameter of Comparison | Detritivores | Decomposers |
|---|---|---|
| **General** | These are organisms which fall under the category of decomposers. | It is a set of organisms which consist of detritivores, scavengers and saprophytes. |
| **Metabolic process** | These organisms do not secrete chemicals to decompose substances. | These organisms secrete chemicals or certain enzymes to decompose dead and decaying organisms. |
| **The method used for degradation** | They degrade different organisms directly through digestion. | These organisms secrete enzymes or chemicals to secrete dead and decaying organisms. |
| **Organic matter consumption** | These consume the organic matter from the substance. | They secrete an enzyme for the digestion of organic matter. |
| **Examples** | Worms, crabs, flies and milipedes. | Bacteria, fungi and insects. |

# MICRO CONSUMERS

# Example of Aquatic Ecosystem

Producers in aquatic ecosystems : blue green algae (cyanobacteria), phytoplankton, diatoms.

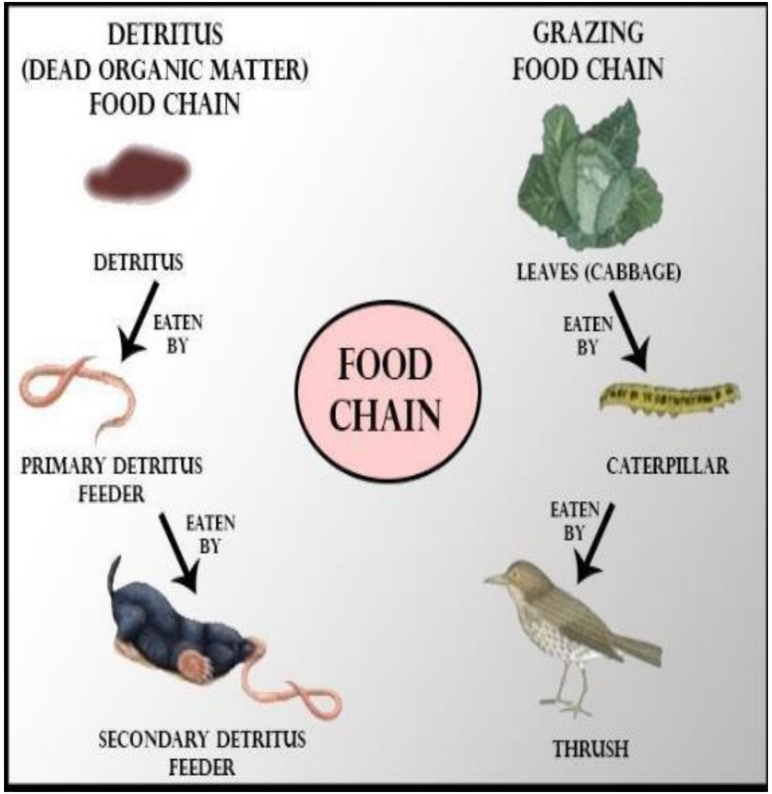
Primary consumers : crustaceans, zooplankton, small fish

Secondary Consumers : Herrings, bigger fish, sharks etc.

# FOOD CHAIN

- A sequence of organisms that feed on one another, form a food chain.
- It is the process of transfer of food energy from green plants (producers) through a series of organisms with repeated eating and being eaten link.
- Arrows in a foodchain represent the flow of energy, through the process of eating.
- Linear: Unidimensional Flow of Energy

# FOOD CHAIN

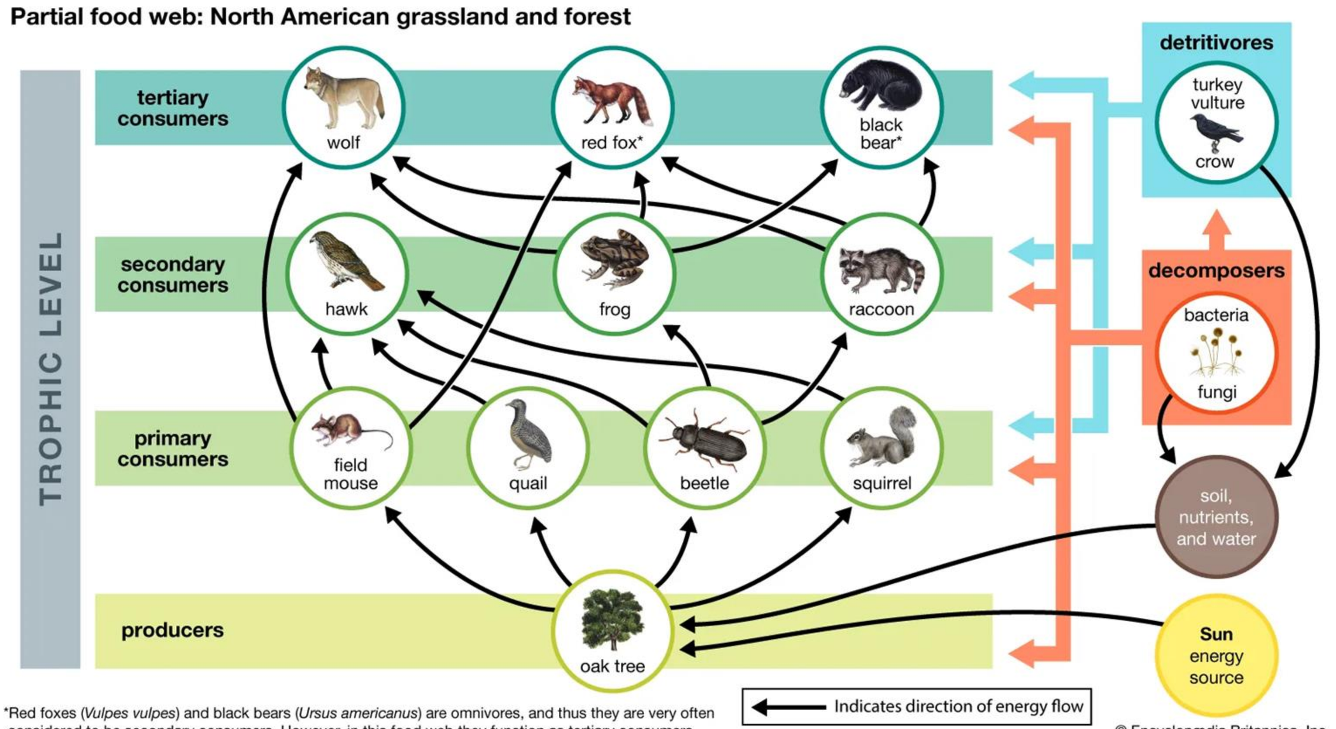


DETRITUS
(DEAD ORGANIC MATTER)
FOOD CHAIN
GRAZING
FOOD CHAIN
DETRITUS
EATEN BY
PRIMARY DETRITUS FEEDER
EATEN BY
SECONDARY DETRITUS FEEDER
FOOD CHAIN
LEAVES (CABBAGE)
EATEN BY
CATERPILLAR
EATEN BY
THRUSH

# FOOD WEB

- A food web is a diagram or model that illustrates the interdependence of various organisms in an ecosystem, showing the flow of energy and nutrients from one organism to another.
- It is a representation of the feeding relationships among different species in an ecosystem.
- **Non Linear:** Multi dimensional Flow of Energy
- **Complexity :** More the number of species, greater the interactions and more complex the food web. These foodwebs are more stable. WHY?

# FOOD WEB




*Red foxes (*Vulpes vulpes*) and black bears (*Ursus americanus*) are omnivores, and thus they are very often considered to be secondary consumers. However, in this food web they function as tertiary consumers.

# FOOD WEB COLLAPSE

- A food web collapse occurs when the interconnected food chain of an ecosystem breakdown or fail to function in a sustainable manner.
- This can happen due to a number of factors such as changes in climate, loss of key species, pollution and overexploitation by humans.
- A food web collapse can have serious consequences on the entire ecosystem, including loss of biodiversity, reduced productivity, and even ecosystem collapse.

# FOOD WEB COLLAPSE



CLIMATE CHANGE

## Climate change could drive coastal food webs to collapse

Benefits of a boost in marine plant growth from increased carbon dioxide will be cancelled out by the increased stress to fish species

NEXT NEWS ›

By Ivan Nagelkerken, Sean Connell , Silvan Goldenberg
Published: Monday 01 May 2017

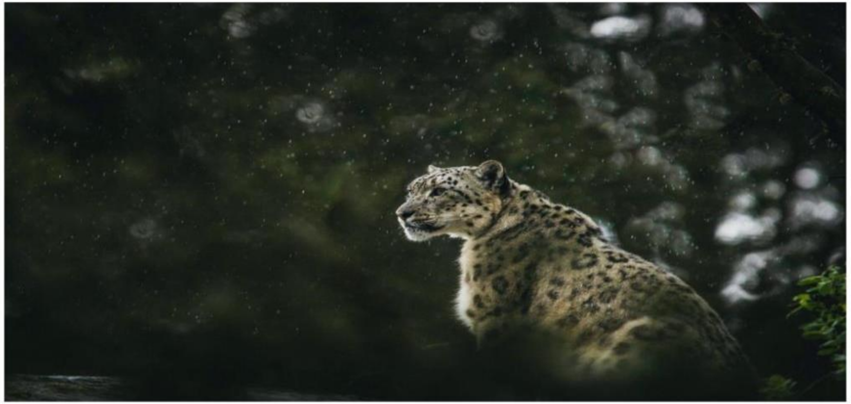
WILDLIFE & BIODIVERSITY

## Children born today will see literally thousands of animals disappear in their lifetime, as global food webs collapse

However, if we manage to achieve a lower carbon-emissions pathway that limits global warming to less than 3°C by the end of this century, we could limit biodiversity loss to “only” 13%

NEXT BLOG ›

By Corey J A Bradshaw, Giovanni Strona
Published: Sunday 18 December 2022



Frida Lannerstrom/Unsplash, CC BY

# ECOLOGICAL PYRAMIDS

- Ecological pyramids are graphical representations of the trophic levels within an ecosystem.
- They show the relative abundance or biomass of different groups of organisms at each trophic level in an ecosystem, and the flow of energy and nutrients through the food chain.

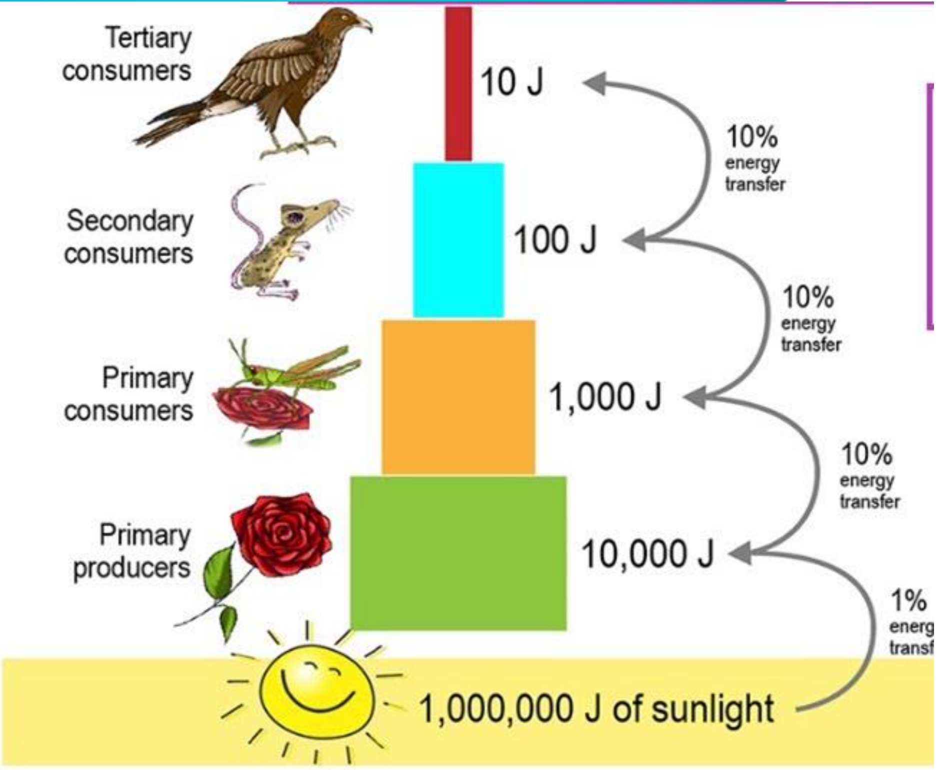
# Law of 10% : given by raymond lindeman

- From one trophic level to another, only biomass will get transferred. However, not all the energy obtained by an organism will get converted to biomass.
- When all energy losses are added, only about 10% of the energy entering one trophic level is available to the next trophic level; because only 10% of the energy obtained is used to make biomass. This is known as the 10% law.

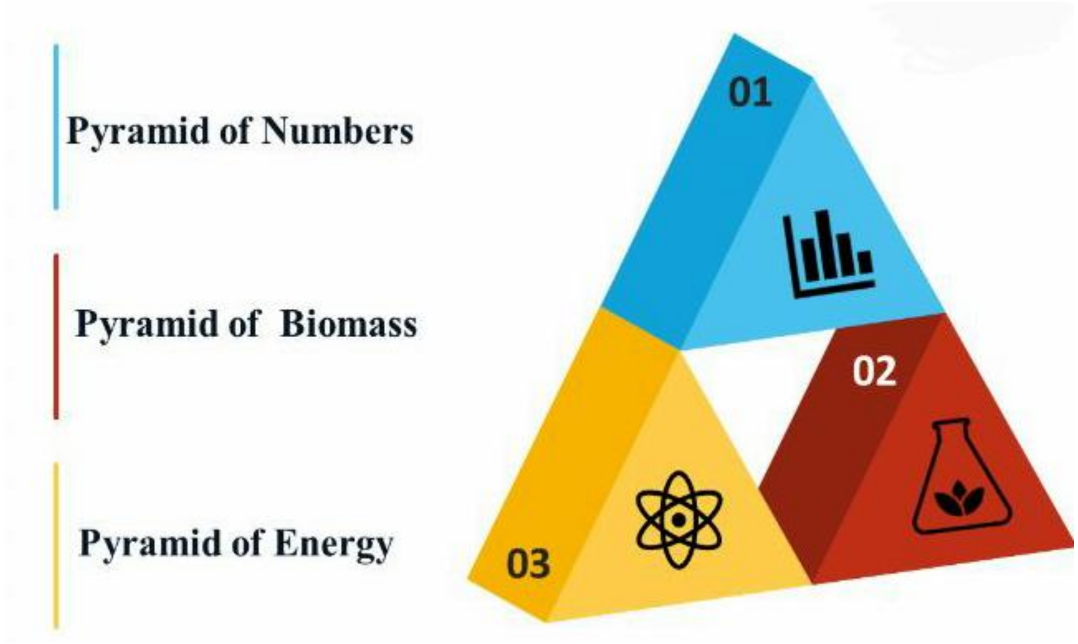
# Law of 10% : given by raymond lindeman

- Because of the 10% law, foodchains have five or less links. Because 90% of energy is lost at each level, the amount of energy available decreases very quickly.
- Most of the energy loss is in production of heat energy and movement from one place to the other.

# Law of 10% : given by raymond lindeman



# ECOLOGICAL PYRAMIDS



# ECOLOGICAL PYRAMIDS



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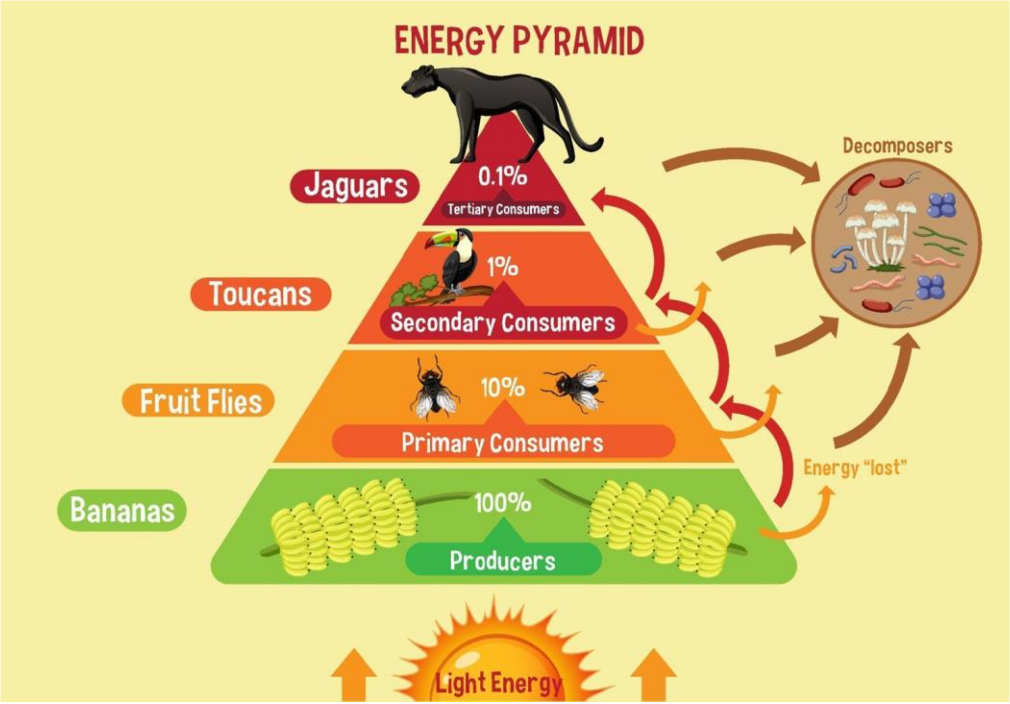
# PYRAMID OF ENERGY

- The pyramid of energy is a graphical representation of the flow of energy through an ecosystem.
- It is always upright.
- As you move up the pyramid, there is less energy available because some is lost in each transfer. (10% Rule)
- This loss of energy is due to the laws of thermodynamics.

# PYRAMID OF ENERGY



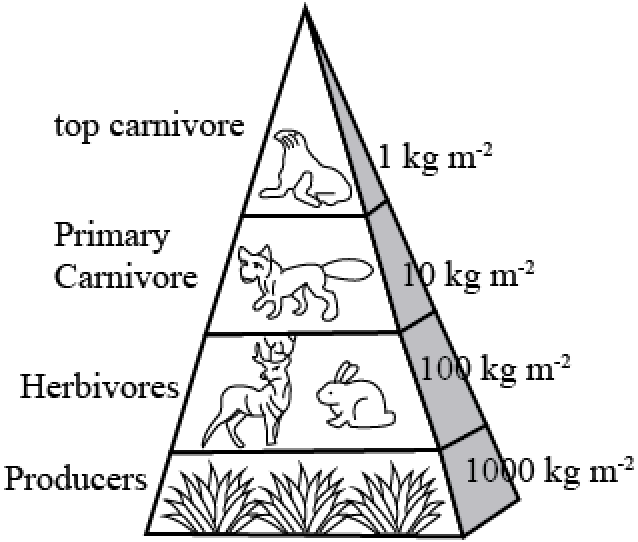
# RELEVANCE OF PYRAMID OF ENERGY

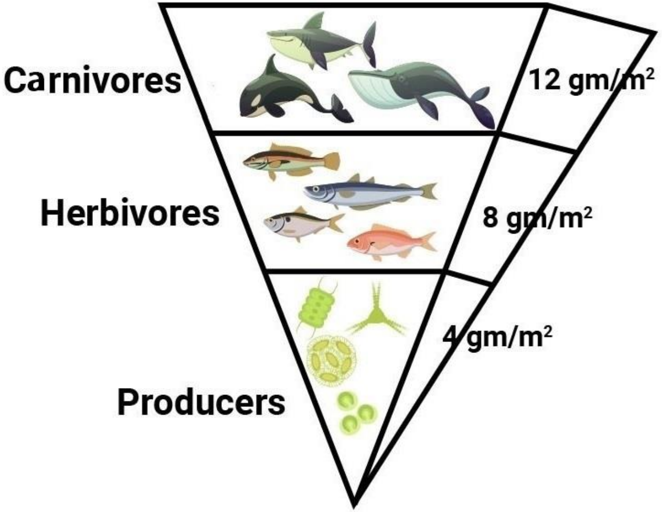
- The food pyramid represents the flow of energy in an ecosystem.
- It illustrates the amount of energy that is transferred from one trophic level to another in an ecosystem.
- The pyramid helps us understand the importance of primary producers and the limits of energy transfer in an ecosystem.
- It is a simple way to visualize the complex interactions between organisms and energy in an ecosystem.

# PYRAMID OF BIOMASS

- The pyramid of biomass is a graphical representation of the amount of living organic matter, or biomass, present in each trophic level of an ecosystem.
- It represents the relative amount of biomass at each level, with the largest biomass at the base of the pyramid and successively smaller biomass at each higher level.
- It is not always a perfect pyramid shape, as it can be affected by factors such as the size and turnover rate of the organisms in each trophic level.

# PYRAMID OF BIOMASS



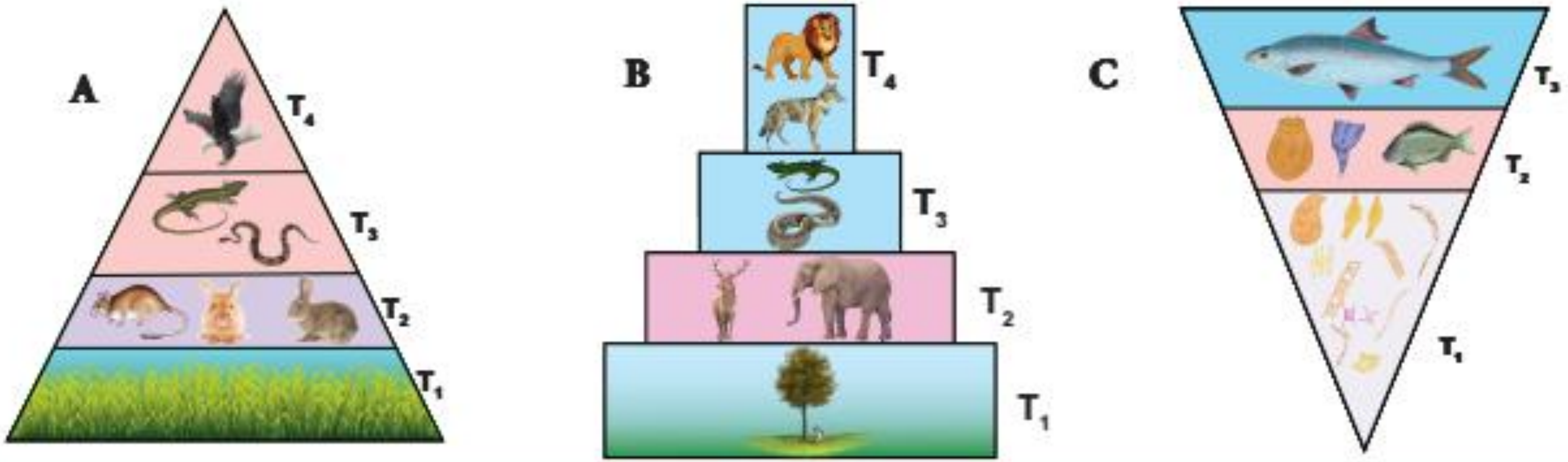




Inverted Pyramid in an Aquatic Ecosystem

# Aquatic pyramid of biomass

- Phytoplankton (2-3 days)
- Zooplankton (7-8 days)
- Small fish (15-20 days)
- Shark (10 years)
- At one particular amount of time, the biomass at the lower trophic levels are lower than that of higher trophic levels. Hence, there is an inverted biomass pyramid.

# PYRAMID OF BIOMASS




$T_1$ - Producers | $T_2$ - Herbivores | $T_3$ - Secondary consumers | $T_4$ - Tertiary consumers

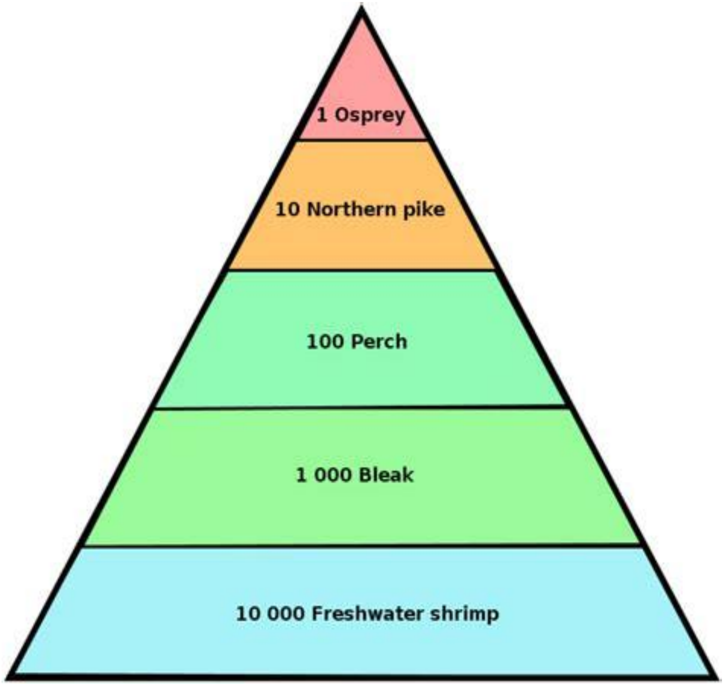
Pyramids of biomass ( dry weight per unit area)in different types of ecosystems.

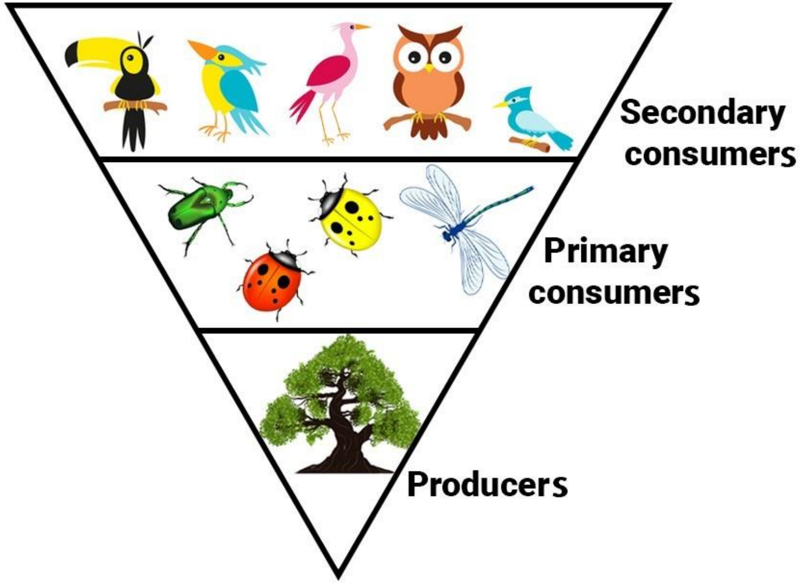
**Upright**-A) Grassland ecosystem B) Forest ecosystem, **Inverted**- C)Pond ecosystem

# PYRAMIDS OF NUMBER

- This pyramid shows the number of organisms at each trophic level in the food chain.
- It indicates the number of individuals at each trophic level, and is often shaped like a true pyramid, with the largest number of organisms at the base and decreasing numbers as you move up the pyramid.
- However, there can be certain exceptions when it is not shaped like a true pyramid.

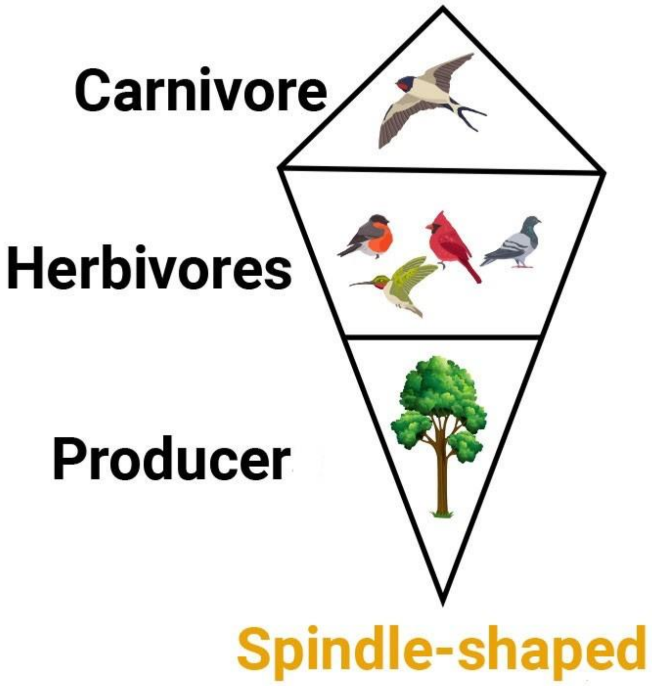
# PYRAMIDS OF NUMBER




Inverted Pyramid of Numbers

# PYRAMIDS OF NUMBER



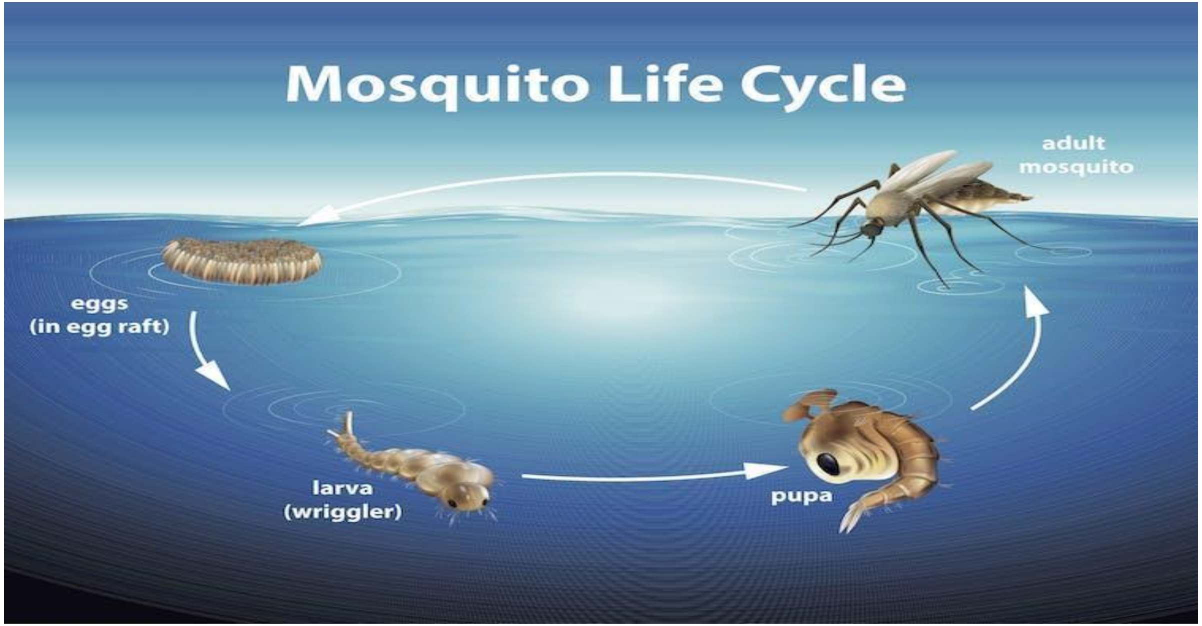
# BIOACCUMULATION

- Bioaccumulation is the gradual accumulation of substances, such as pesticides or other chemicals, in an organism.
- It occurs when an organism absorbs a substance at a rate faster than that at which the substance is lost or eliminated by catabolism and excretion.

# Accumulation in Food Chain (Down to Earth)

- Microplastics are getting into mosquitoes and contaminating new food chains
- According to a research, there was evidence of beads in all the life stages of Mosquitoes, although the numbers went down as the animals developed.
- Any flying insect that spends part of its life in water can become a carrier of plastic pollution thus resulting in Biomagnification at higher trophics.

# Accumulation in Food Chain (Down to Earth)



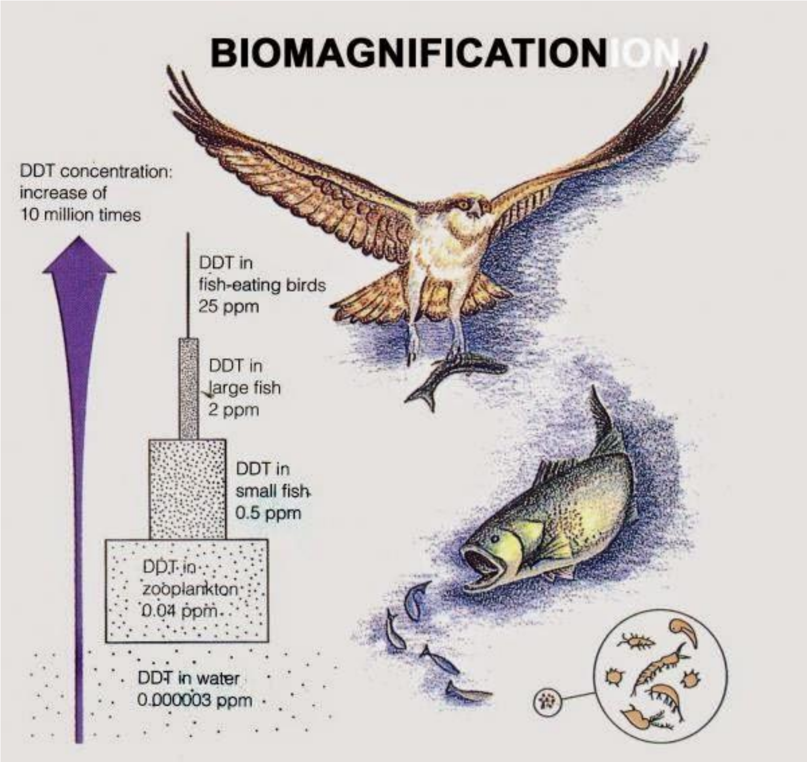
# Biomagnification vs Bioaccumulation

- Bioaccumulation takes place in a single organism over the span of its life, resulting in a higher concentration in older individuals. Biomagnification takes place as chemicals transfer from lower trophic levels to higher trophic levels within a food web, resulting in a higher concentration in apex predators.
- When DDT enters aquatic bodies, it gets build up in the body of fishes and this is known as bioaccumulation. When fishes are eaten by animals of higher trophic levels, concentration of DDT is increased at each successive trophic level and this is known as biomagnification.

# BIOMAGNIFICATION

- Biomagnification, also known as bioamplification or biological magnification, is the increase in concentration of a substance, e.g a pesticide, in the tissues of organisms at successively higher levels in a food chain.
- This increase can occur as a result of:
- Persistence – where the substance cannot be broken down by environmental processes
- Food chain energetics – where the substance's concentration increases progressively as it moves up a food chain
- Low or non-existent rate of internal degradation or excretion of the substance – mainly due to water-insolubility

# SUBSTANCES THAT BIOMAGNIFY



# SUBSTANCES THAT BIOMAGNIFY

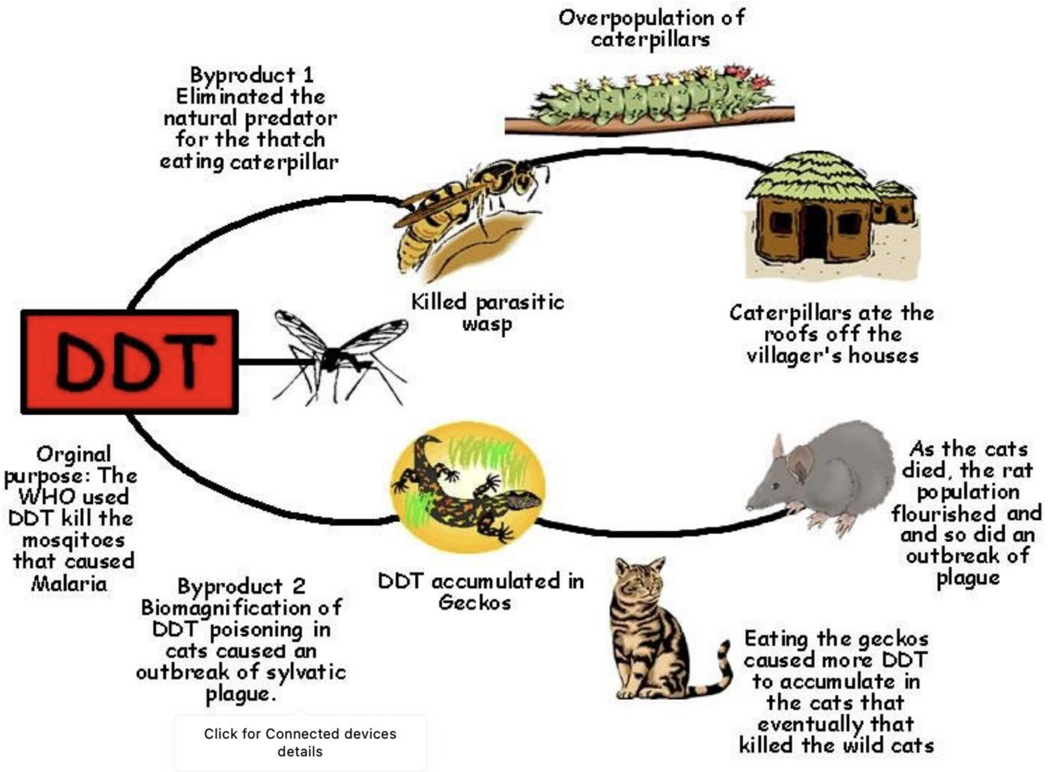
- In order for biomagnifcation to occur, the pollutant must be:

  **long-lived, mobile, soluble in fats, non biodegradable**
- **Examples of Substances:**
  - Chlorinated hydrocarbons (Organochlorines)
  - Inorganic compounds like methylmercury or heavy metals
  - Persistent organic pollutants

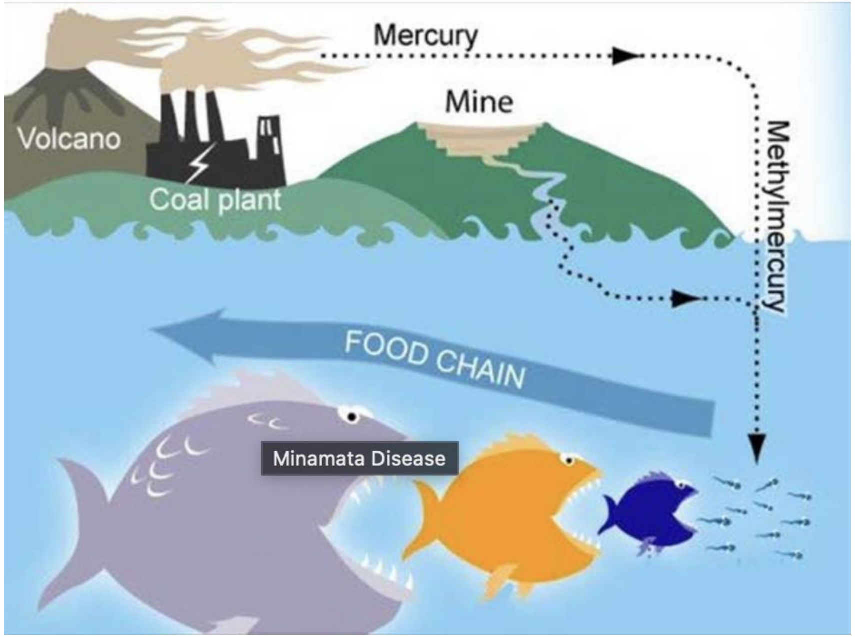
# Examples of Biomagnification

- DDT : used to kill mosquitoes, added up in bald eagle populations leading to their egg shells being crushed. This was written about by Rachel Carson in her book the silent spring.
- It rained cats in borneo! : explained in class.
- Minamata Disease : Mercury bioaccumulation in shellfish in minamata bay, eaten by the local populace leading to mercury poisoning.
- Diclofenac and the indian vulture crisis.

# Examples of Biomagnification



# Examples of Biomagnification



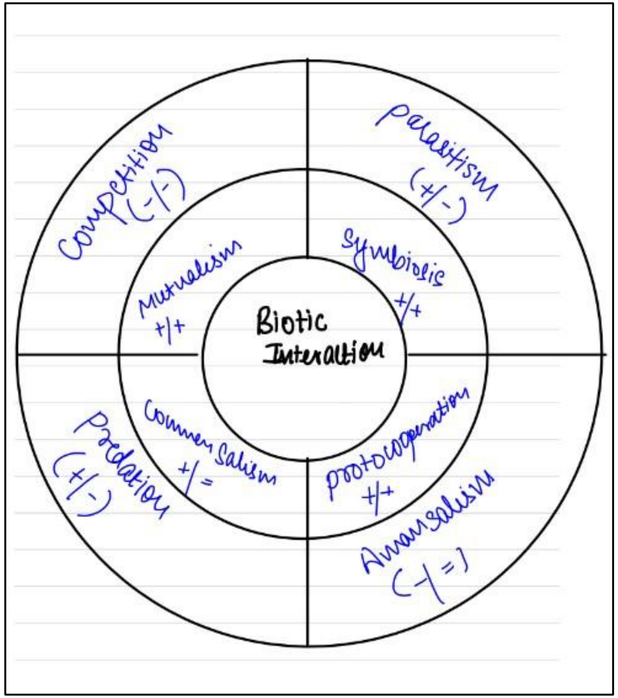

# Biomagnification in Vultures



- Traces of diclofenac and its derivative compounds have been found in the carcasses of vultures across India and its neighboring countries, and it is known that the biomagnification of diclofenac from the consumption of infected domestic animal carcasses contributes to vulture mortality.

# BIOTIC INTERACTIONS

- Biotic interactions are the interactions between different living organisms in an ecosystem.
- These interactions can be positive, negative, or neutral, and they can affect the distribution and abundance of species, the structure of communities, and the functioning of ecosystems.



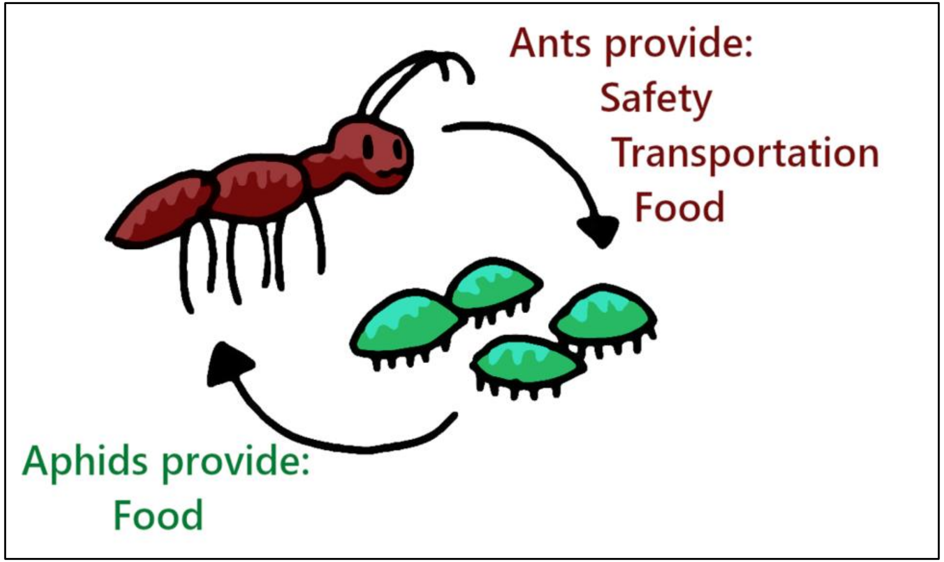
# BIOTIC INTERACTIONS

| SN. NO. | TYPES OF INTERATION | SPECIES 1 | SPECIES 2 | GENERAL NATUE OF INTERACTION | EXAMPLES |
|---|---|---|---|---|---|
| 1 | Amensalism | – | 0 | The most powerful animal or large organisms inhibits the growth of other lower organisms | Cat and Rat |
| 2 | Mutualism | + | + | Interaction favorable to both and obligatory | Between crocodile and bird |
| 3 | Commensalism | + | 0 | Population 1, the commensal benefits, while 2 the host is not affected | Sucker fish on shark |
| 4 | Competition | – | – | Direct inhibition of each species by the other | Birds compete with squirrels for nuts and seeds. |
| 5 | Parasitism | + | – | Population 1, the parasite, generally smaller than 2, the host | *Ascaris* and tapeworm in human digestive tract. |
| 6 | Predation | + | – | Population 1, the predator, generally larger than 2, the prey | Lion predatory on deer |

# MUTUALISM

- It is the type of interaction where both species benefit from the interaction.
- It highlights the interdependence of species in an ecosystem. By working together, species can enhance their chances of survival and reproduction, and help to maintain the balance of the ecosystem.

# MUTUALISM



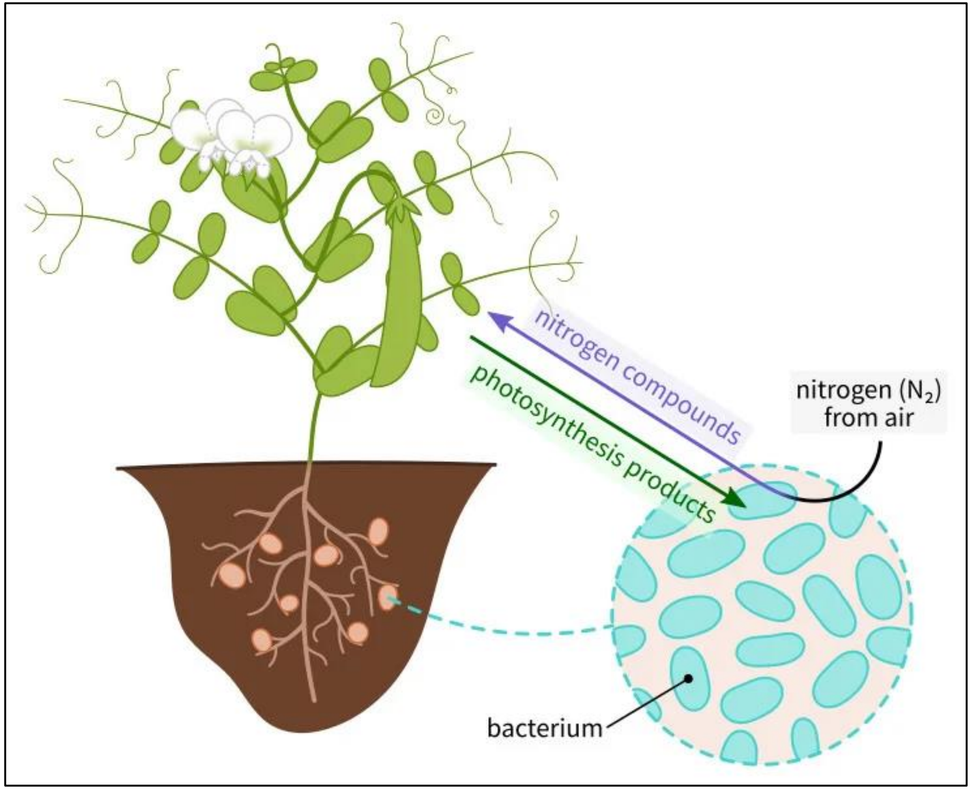
Ants provide:
Safety
Transportation
Food
Aphids provide:
Food

# SYMBIOSIS



- Symbiosis is a type of biotic interaction in which two or more different species live in close association with each other.
- The term "symbiosis" was coined by Anton de Bary in 1879 and is derived from the Greek words "syn" (together) and "biosis" (living).
- Symbiosis can take many different forms, and not all symbiotic relationships are mutualistic.
- Some symbiotic relationships are parasitic, in which one species benefits at the expense of the other, while others are commensal, in which one species benefits without affecting the other.

# SYMBIOSIS



nitrogen compounds
photosynthesis products
nitrogen ($N_2$) from air
bacterium

# PROTO COOPERATION

- Proto-cooperation is a type of interaction between two species that is not yet fully mutualistic but represents the early stages of mutualism.
- In proto-cooperation, two or more species interact in a way that benefits both, but the relationship is not yet fully dependent on each other.
- Over time, the relationship may evolve into a more symbiotic and mutualistic one.

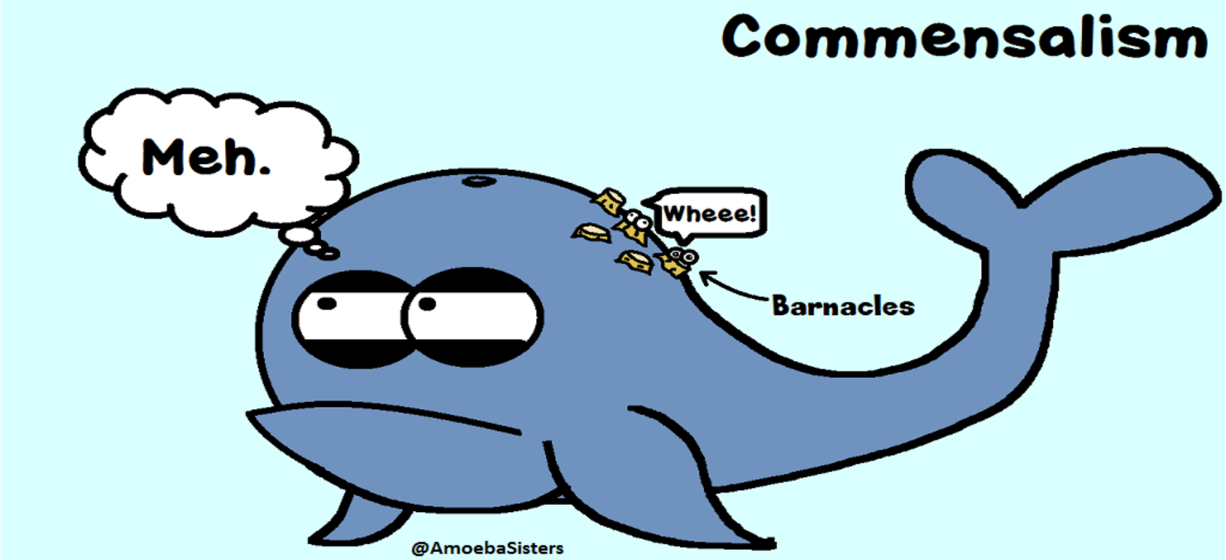
# PROTO COOPERATION

- In it, Interaction is temporary. Once, the interacting organisms get the advantage, they separate.
- Survival is not an issue.
- Example: Plants and pollinators.
- The pollinators may feed on the nectar of the plants, but they also inadvertently transfer pollen from one plant to another, benefiting the plants' reproductive success.

# COMMENSALISM

- This defines the interaction in which two or more species are mutually associated in activities centering on food and one species at least, derives benefit from the association while the other associates are neither benefited nor harmed.

# COMMENSALISM



# PREDATION

- It is a form of interaction, where one animal kills another animal for food.
- Specialized predators are those adapted to hunt only a few specific species.
- Lion and deer exhibit predator – prey relationship, where the Lion is the predator and the deer is the prey.
- This type of interaction helps in the transfer of energy up the trophic levels and is an essential strategy in population regulation.

# PREDATION

# PARASITISM

- It is a kind of harmful interaction between two species, wherein one species is the ‘parasite’ and the other its ‘host’.
- The parasite benefits at the expense of the host.
- A parasite derives shelter, food and protection from the host.
- Parasites exhibit adaptations to exploit their hosts.

# PARASITISM

# PARASITISM

- The parasites may be:
  - viral parasites (plant/ animal viruses),
  - microbial parasites (e.g., bacteria / protozoa / fungi),
  - phyto parasites (plant parasites)
  - zooparasites (animal parasites such as Platyhelminthes, nematodes, arthropods).
- Parasites may inhibit or attach to the surface of the host (Ectoparasites - Head lice, Leech) or live within the body of the host (endoparasites – ascaris, tapeworm).

# COMPETITION

- It refers to the type of interaction in which individuals of a species or members of different species vie for limited availability of food, water, nesting space, cover, mates or other resources.
- When resources are in more than adequate to meet the demands of the organisms seeking them, competition does not occur, but when inadequate to satisfy the need of the organisms seeking them, the weakest, least adapted, or least aggressive individuals are often forced to face challenges.

# INTERSPECIFIC COMPETITION

- Interspecific competition is a type of interaction in which two or more species compete for a limited resource, such as food, water, or habitat.
- This competition occurs between different species that share the same resource requirements.
- For example, lions and hyenas may compete for access to prey species, or two species of trees may compete for sunlight and nutrients.

# INTERSPECIFIC COMPETITION

# INTRASPECIFIC COMPETITION

- Intraspecific competition, on the other hand, is a type of interaction in which individuals of the same species compete for resources.
- This competition occurs within a species, and it can be intense, especially when resources are scarce.
- For example, plants may compete for water and nutrients in the soil, or animals may compete for mates or nesting sites.

# AMENSALISM

- This is the ecological interaction in which an individual species harm another without obtaining benefit.
- In anmensalism, the species that is negatively affected is often referred to as the "victim" species.
- The negative effect can occur in several ways, such as through the release of chemicals or other substances that are toxic or inhibitory to the victim species, or through physical interference with the victim's ability to access resources.

# AMENSALISM

- Example: A large tree shades a small plant, retarding the growth of the small plant. The small plant has no effect on the large tree.

# Test of 8 Questions!

## Cut off : 6/8

# Q.1 (2015)

Q.) Which one of the following is the best description of the term 'ecosystem'?

(a) A community of organisms interacting with one another

(b) That part of the Earth which is inhabited by living organisms

(c) A community of organisms together with the environment in which they live

(d) The flora and fauna of a geographical area

## Q.2 (2013)

Q.) Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms?

(a) Ecotone

(b) Ecological niche

(c) Habitat

(d) Home range

## Q.3 (2014)

Q.) Which one of the following is the correct sequence of a food chain?

(a) Diatoms-Crustaceans-Herrings

(b) Crustaceans-Diatoms-Herrings

(c) Diatoms-Herrings-Crustaceans

(d) Crustaceans-Herrings-Diatoms

# Q.4 (2013)

Q.) With reference to food chains in ecosystems, consider the following statements

1. A food chain illustrates the order in which a chain of organisms feed upon each other.
2. Food chains are found within the populations of a species.
3. A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is / are correct?

(a) 1 only

(b) 1 and 2 only

(c) 1, 2 and 3

(d) None

## Q.5 (2013)

Q.) With reference to the food chains in ecosystems, which of the following kinds of organism is / are known as decomposer organism/organisms?

1. Virus
2. Fungi
3. Bacteria

Select the correct answer using the codes given below.

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Q.) Which of the following are detritivores?

1. Earthworms

2. Jellyfish

3. Millipedes

3. Seahorse

4. Woodlice

Select the correct answer using the code given below

(a) 1, 2 and 4 only

(b) 2, 3, 4 and 4 only

(c) 1, 3 and 5 only

(d) 1, 2, 3, 4 and 5

# Q.7 (2021)

Q.) Consider the following kinds of organisms:

1. Copepods
2. Cyanobacteria
3. Diatoms
4. Foraminifera

Which of the above are primary producers in the food chains of oceans?

A. 1 and 2

B. 2 and 3

C. 3 and 4

D. 1 and 4

## Q.8 (2021)



Q.) Which of the following have species that can establish symbiotic relationship with other organisms?

1. Cnidarians
2. Fungi
3. Protozoa

Select the correct answer using the code given below.

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

# Congratulations on giving the test!

# Let’s Discuss The Answers.

## Q.1 (2015)

Q.) Which one of the following is the best description of the term 'ecosystem'?

(a) A community of organisms interacting with one another

(b) That part of the Earth which is inhabited by living organisms

**(c) A community of organisms together with the environment in which they live**

(d) The flora and fauna of a geographical area

# Answer: C

- The term ‘Ecosystem’ was first used by A G Tansley in 1935 who defined an ecosystem as ‘a particular category of physical systems, consisting of organisms and inorganic components in a relatively stable equilibrium, open and of various sizes and kinds’.
- Thus the ecosystem includes both the organisms as well as their environment.

## Q.2 (2013)

Q.) Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms?

(a) Ecotone

**(b) Ecological niche**

(c) Habitat

(d) Home range

# Answer: B

- In ecology, a niche is a term describing the relational position of a species or population in an ecosystem.
- More formally, the niche includes how a population responds to the abundance of its resources and enemies (e. g., by growing when resources are abundant, and predators, parasites and pathogens are scarce) and how it affects those same factors (e. g., by reducing the abundance of resources through consumption and contributing to the population growth of enemies by falling prey to them).
- Gause’s competitive exclusion principle says that those species having identical requirements cannot occupy the same ‘niche’ indefinitely.

## Q.3 (2014)

Q.) Which one of the following is the correct sequence of a food chain?

**(a) Diatoms-Crustaceans-Herrings**

(b) Crustaceans-Diatoms-Herrings

(c) Diatoms-Herrings-Crustaceans

(d) Crustaceans-Herrings-Diatoms

# Answer: A

- Diatoms are a major group of algae and are among the most common types of phytoplankton, so primary producers/autotrophs.
- Crustaceans as consumers/heterotrophs while Herrings (a fish) feed on Crustaceans.

## Q.4 (2013)

Q.) With reference to food chains in ecosystems, consider the following statements

1. A food chain illustrates the order in which a chain of organisms feed upon each other.
2. Food chains are found within the populations of a species.
3. A food chain illustrates the numbers of each organism which are eaten by others.

Which of the statements given above is / are correct?

**(a) 1 only** (b) 1 and 2 only

(c) 1, 2 and 3 (d) None

# Answer: A

- Food chain, in ecology, the sequence of transfers of matter and energy in the form of food from organism to organism.
- Food chains are not found within the populations of 'a' species, because technically, the food chain is the sequence of organisms through which the energy flows.
- Food chains intertwine locally into a food web because most organisms consume more than one type of animal or plant.

## Q.5 (2013)

Q.) With reference to the food chains in ecosystems, which of the following kinds of organism is / are known as decomposer organism/organisms?

1. Virus
2. Fungi
3. Bacteria

Select the correct answer using the codes given below.

(a) 1 only

**(b) 2 and 3 only**

(c) 1 and 3 only

(d) 1, 2 and 3

# Answer: B

- Fungi and Bacteria are decomposers.
- They break down organic matter into simple inorganic substances.
- Virus represents dormant life. They are metabolically inactive as long as they are outside a host body.
- They are not decomposers. They invade host cells and use their nucleus (DNA machinery) to carry out their life processes.

# Q.6 (2021)

Q.) Which of the following are detritivores?

1. Earthworms

2. Jellyfish

3. Millipedes

3. Seahorse

4. Woodlice

Select the correct answer using the code given below

(a) 1, 2 and 4 only

(b) 2, 3, 4 and 4 only

**(c) 1, 3 and 5 only**

(d) 1, 2, 3, 4 and 5

# Answer: C

- Detritivores are heterotrophs that obtain nutrients by consuming detritus.
- There are many kinds of invertebrates, vertebrates and plants that carry out coprophagy.
- By doing so, all these detritivores contribute to decomposition and the nutrient cycles.
- Earthworm, Millipedes and Woodlice are detritivores.

Q.) Consider the following kinds of organisms:

1. Copepods
2. Cyanobacteria
3. Diatoms
4. Foraminifera

Which of the above are primary producers in the food chains of oceans?

A. 1 and 2

**B. 2 and 3**

C. 3 and 4

D. 1 and 4

# Answer: B

- Autotrophs or primary producers are organisms that acquire their energy from sunlight and materials from non-living sources. Copepods are a group of small crustaceans found in nearly every freshwater and saltwater habitat. Copepods are major secondary producers in the World Ocean.
- They represent an important link between phytoplankton, microzooplankton and higher trophic levels such as fish. They are an important source of food for many fish species but also a significant producer of detritus. Hence, option 1 is not correct.

## Q.8 (2021)

Q.) Which of the following have species that can establish symbiotic relationship with other organisms?

1. Cnidarians
2. Fungi
3. Protozoa

Select the correct answer using the code given below.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

**D. 1, 2 and 3**

# Answer: D

- Cnidarian, also called coelenterate are mostly marine animals. They include the corals, hydras, jellyfish, Portuguese menof-war, sea anemones, sea pens, sea whips, and sea fans. The relationship between cnidarians and dinoflagellate algae is termed as "symbiotic", because both the animal host and the algae are benefiting from the association.
- It is a mutualistic interaction Fungi have several mutualistic relationships with other organisms. In mutualism, both organisms benefit from the relationship. Two common mutualistic relationships involving fungi are mycorrhiza and lichen.
- Termites have a mutualistic relationship with protozoa that live in the insect's gut. The termite benefits from the ability of bacterial symbionts within the protozoa to digest cellulose.

- Cyanobacteria, also called blue-green algae, are microscopic organisms found naturally in all types of water. Cyanobacteria are important primary producers and form a part of the phytoplankton. They may also form biofilms and mats (benthic cyanobacteria). Hence option 2 is correct.
- Diatoms are photosynthesising algae, they have a siliceous skeleton (frustule) and are found in almost every aquatic environment including fresh and marine waters. Diatoms are one of the major primary producers in the ocean, responsible annually for ~20% of photosynthetically fixed $CO_2$ on Earth. Hence option 3 is correct.
- Foraminifera are single-celled organisms, members of a phylum or class of amoeboid protists characterized by streaming granular ectoplasm for catching food and other uses. Hence, option 4 is not correct.